Answers & Explanations

the catapult papers

FOR THE GROWN-UPS

Tests 1-10

2026 edition

Notes for parents

The answers booklet for the Catapult Papers includes explanations for every answer.

Each explanation is brief, however, as they are limited to two lines per question.You should nevertheless discover a broad range of helpful insights and tips that could, in turn, be used to explain mistakes to your child. Should you find yourself requiring further explanations for any of the questions in this series, please do not hesitate to contact me (Trevor) at info@catapult-tuition.com.

When reading the explanations throughout this series, you will come across a range of strategies. For example, when it comes to spellings challenges, I will occasionally point out regular spelling rules and patterns that can be transferred across many other words. That said, we are also aware that the English language has many inconsistencies and confusing spellings; this means that it is necessary, on occasion, to be more creative (even childlike) in creating 'memory hooks' for certain words.

You will also discover that, for certain Maths questions, I would encourage children to test out each option (this is one of the benefits of multiple-choice questions) but, on other occasions, I will point out fast 'tricks' that could help children come to the answers quickly. There is rarely one 'correct' way to solve a Maths question. It is up to you to discern which approaches work best for your child.

It should also be emphasised strongly that *this booklet is written for adults*. While you can hand this booklet over to your child for self-marking purposes, please do not encourage them to read the explanations. True, there will be many explanations that children will readily understand but there are also many technical terms used that are intended for adults (particularly in the English sections).

Finally, please do not allow your child (or yourself!) to become disheartened by 'low' scores, especially in early tests. As you will no doubt discover, the Transfer Tests often include questions that we adults would answer incorrectly. The idea of practice is to slowly but surely grow in knowledge, skills and understanding. Indeed, by the end of the Transfer Test season, parents who are new to the process often find they have learned new and interesting things. That doesn't mean these months are going to be fun... but we should take what we can get!

TEST 1 ANSWERS >>>

Test 1 English Answers

1	A	*It's all in a <u>day's</u> work...* The *work* belongs to the *day*. A **possessive apostrophe** should be placed after the spelling of *day* because the work belongs to a (singular) day. Compare this to the apostrophe at the end of *others'* (the *lives* belong to *others*).
2	B	*Frank was often warned by his <u>m</u>um...* In this sentence, *mum* is **not a proper noun**. Names like *Mum* and *Dad* are only capitalised when intended as their 'actual name'. Compare these: I told <u>Mum</u> the news / I told <u>my mum</u> the news.
3	N	The two words using apostrophes are contractions and are used properly. In both cases, the apostrophe replaces the letter o from the word *not* (was not = *wasn't*; were not = *weren't*).
4	B	The **comma** must come <u>before</u> the closing speech mark and not after it. Closing speech marks must always be <u>preceded</u> by a punctuation mark.
5	A	The phrase *some people say* has been inserted into the sentence (it is a dependent clause). In these cases, the phrase must have **a comma** <u>before and after</u> it. This means that a comma must come <u>before</u> the word *some*.

6	C	*Whenever Lauren and Jane* **lose**, *they <u>become</u> difficult people to be around!* The sentence is in the present tense, which means we must choose either *loose* or *lose*. *Loose* means to 'set free' and rhymes with 'goose'.
7	D	*A huge meteor came plummeting* **towards** *the planet at a phenomenal speed.* All of the options would make sentences that sound grammatical, but *towards* is the most sensible preposition to use here: *plummeting* tells us of the downward direction.
8	E	*Janet <u>had</u> never* **sung** *with this particular choir before.* The key word is *had*. When a verb is preceded by *had* (or *has*, *have*, etc.) it becomes a past participle. Compare these: Janet never **sang** / Janet **<u>had</u>** never **sung**.
9	B	**Despite** *my protests, Mum made me tidy my room. Despite my protests* works the same way as 'Even though I protested...'.
10	B	**Although** *it wasn't our fault, we all had to take the blame for what someone in the other class did.* A conjunction must be chosen that shows a contrast: the children suffered a consequence <u>even though</u> it wasn't their fault.

11	B	The spelling should be **occa<u>s</u>ionally**. There is only one s. If it were to have a double s then we would expect the *-assion* to sound like <u>*passion*</u>. There is a softer z sound, however, which hints at the use of only one s (we hear this sound in *A<u>s</u>ia*).
12	D	The spelling should be **apostroph<u>e</u>s**. This is an unusual spelling and should be learned (especially as the word *trophy* / *trophies* ends with *-y* / *-ies*). The only other common word that ends like *apostr<u>ophe</u>* is *catastr<u>ophe</u>*.
13	D	The spelling should be **lib<u>r</u>ary**. This is often misspelled because we say the word quickly. It may help to exaggerate the '*RAR*', making a sound similar to a loud roar – a noise we should never make in a library!
14	A	The spelling should be **atten<u>d</u>ance**. There is no easy way to know which words end with *-ance* and which end with *-ence*. It may be best to find a silly but memorable way to remember this spelling (perhaps by using the word *dance*).
15	N	There are no spelling mistakes. <u>*P*</u>*atiently* is often misspelled; the opening syllable sounds a little like 'pay'.

16	C	The watcher's behaviour adds **tension** to the mood. It moves *in the shadows* (line 2); it follows *at a distance* (lines 6-7); it approaches her room (lines 8-9). As we don't see it or hear from it, this also adds **mystery** to the mood.
17	C	See line 15: *...she seemed <u>hesitant</u> to believe her*. Mrs Fairfax is <u>hesitating</u>. She is not prepared to immediately believe Charlotte, but she does wonder. Even if children don't know the meaning of *reluctant*, process of elimination should help.
18	A	*Motherly* is describing the <u>noun</u> *housekeeper*. **Adjectives** are words that describe nouns.
19	E	Lines 48-51 help describe the plan in clearer detail, and therefore help us understand the meaning of words like *oblivious*. Charlotte <u>pretends</u> to admire the flowers (line 49). She is **acting as if she's not concerned** about the watcher.
20	C	See line 5: *...and put an end to whatever* **nefarious** [wicked] *plot they were hatching...* A<u>*miss*</u> means '<u>miss</u>ing/not right/out of place'. *In vain* means 'fruitless'. *Confide* is a verb that means 'entrust'. *Grave* means 'serious'.
21	B	Option A is clearly false. There is no evidence that Charlotte enjoys acting, so option C is not a good choice. Option D isn't true as she only pretended to admire the flowers. Option E doesn't explain her fear.
22	C	Lines 53-55 hold the clue: Charlotte removes the toy from the watcher's <u>*mouth*</u>! She is also confident while she does so. It is presumably a dog that has been following her. We may assume Charlotte wouldn't do this to a human!

23	crumbs	The answer is on lines 4-5. Accept longer answers but it must include **crumbs**.
24	Thornfield	See lines 13, 17, 23 and 26.
25	persisted	In these types of questions, it usually works to swap the word in the question with a word in the passage to see if it sounds correct. In this case, we must find another past tense verb.
26	28	*The weight of her secret pressed down on her <u>like a lead blanket</u>...* Similes are phrases that compare things because of something they have in common. They use the words *as* or *like*.
27	increasingly	Many adverbs have *-ly* as an ending, which makes them easier to find. What kids may not know, however, is that adverbs can describe adjectives: *...increasingly <u>concerned</u>...*.
28	noun	It is vital that kids check the words within the sentences. In line 1, *glimpse* is a noun (even though it is often a verb). We know this because it is preceded by the word *a*. 'A glimpse' is a thing.

29	E	Two calculations are required: 117 - 19 (98), then add the answer to 117. 117 + 98 = **215**
30	B	A right angle measures exactly 90˚, so A and B will add together to make this total. If B is 35˚ then A must be **55˚** (90˚ - 35˚).
31	A	If ¾ of the shape is shaded then <u>¼ is not</u>. It is therefore easier to count the unshaded squares. Straight away, in **Shape A**, we can see that there are 8 unshaded squares, but 28 squares (4x7) in total. 8 is not one-quarter of 28.
32	D	The factors of 12 are <u>1</u> and 12, <u>2</u> and 6, 3 and <u>4</u>. The factors of 16 are <u>1</u> and 16, <u>2</u> and 8, and <u>4</u>. In other words, these two numbers share **three** factors in common: 1, 2 and 4.
33	B	As the front and back faces are squares, the vertical and horizontal measurements must be the same. Three measurements are multiplied together to get the volume: □ x □ x 10cm = 90cm³ (the two missing numbers are the same). It has to be **3cm**.
34	D	6 tens = <u>60</u>; 3 ten thousands = <u>30000</u>; 8 tenths = <u>0.8</u>. 60 + 30000 + 0.8 = **30060.8**
35	D	75% is the same as **¾**. To find three-quarters of a number, we divide by four then multiply the answer by three. 500 ÷ 4 = 125 125 x 3 = **375**
36	C	If there are four performances, there are only <u>three breaks</u> (so 15 minutes of breaks altogether). 48 mins + 15 mins = 1 hr 3 mins. 6.45pm + 1 hr 3 mins = **7.48pm**.
37	C	£1.65 x 2 = £3.30, then add 89p to reach £4.19. There will be <u>81p change</u> from £5. The fewest number of coins is **four**: 50p, 20p, 10p and 1p.
38	E	For option A, 1˚+1˚+1˚ could be used as a test. For option B, 30˚+30˚+30˚ works. For option C, 40˚+40˚+40˚ works. For option D, 70˚+70˚+70˚ works. But you can't make **360˚** by adding three acute angles together (they are less than 90˚ each).
39	B	45˚ is half of a right angle (90˚... or one-quarter of the pie chart). The shape of the *Audi* segment might give away that it is one-quarter of the chart (18 is **¼** of 72). The car with the 45˚ segment will have <u>half Audi's number</u>: 18÷2=<u>9</u>... It is **BMW**.
40	C	The distance to West City is a fraction of the distance to East City, and is therefore shorter. Divide 43.4 by 7 (6.2), then multiply the answer by 5. The answer is **31km**.
41	B	The numbers have to be <u>both</u> square <u>and</u> divisible by 4. Although 72 is divisible by 4, it is not a square number. Only <u>16</u> and <u>100</u> meet the conditions.
42	D	The 'cube' from Net A would either have no base or lid. Nets D & E won't work because of their 2x2 sections. **Net C** is the most common arrangement that works (1/4/1). 2x3 and 3x2 'stairs' arrangements also work, which makes **Net B** possible.
43	B	Always read the whole number before the decimal part! It is either option B or E. Option B has the digit 4 in the hundredths place, which is less than option E (its digit 4 is in the tenths place, making it bigger). The answer is **201.04**.
44	A	Kids can imagine that the image was once a full rectangle (before the top-right corner was removed). This would be a 20cm by 5cm rectangle (100cm²). The 'missing piece' measures 2cm by 5cm (10cm²). Subtract this from 100cm² to get **90cm²**.
45	B	Mr Jackson arrives at his office at 8.30am. He remains there until 12.30pm. So far, this is <u>4 hours</u> at the office. If he arrives home at 5.30pm, and his journey took 45 minutes, then he left the office at 4.45pm. This is another <u>3 hours 30 minutes</u> at work.
46	B	Each interval on the thermometer scale represents 2˚C. The temperature in Florida is therefore 16˚C. Subtract 20˚C from this to get **-4˚C**.
47	D	The prime numbers between twenty and thirty are <u>23 and 29</u> (21 and 27 are multiples of three). The sum of these numbers is 52. Divide this by two (the number of numbers you added) to get the mean: **26**.
48	C	Each option should be tested until something works. The brother's amount should be £12 less than Martha's, after which their combined amount should make £90. If his amount is **£39**, Martha's will be £51. These two amounts add to make £90.
49	A	Write **⅚** beside each option. The only one that will work will have a numerator and denominator equally as great as 5 and 6. The answer is **40/48** because 40 is eight times greater than 5, and 48 is eight times greater than 6.
50	E	Shape 1 is a <u>2x5</u> arrangement. Shape 2 is a <u>3x6</u> arrangement. Shape 3 is a <u>4x7</u> arrangement. The respective numbers are increasing by one. Shape 4 must be a <u>5x8</u> arrangement, which makes **40** squares.

51	44 cm²	Areas of triangles are calculated by multiplying the base by the height, then halving the answer. Sometimes it is quicker to halve one of the measurements at the start and then multiply (4x11).
52	1310	<u>1.31</u> is the largest of the numbers. When multiplied by 1000, this makes **1310**.
53	28 cm	The area of the square is 144cm². The area of the rectangle is one-third of this, so must be 48cm². We must find its perimeter. The missing measurement is 8cm (48 ÷ 6). 8 + 8 + 6 + 6 = **28**.
54	17 edges	All of the edges are visible in the images because the shapes are transparent: kids can count the lines if they can't remember the facts. The pyramid has <u>8 edges</u>, the prism has <u>9 edges</u>.
55	28 visitors	Each interval on the y axis of the graph is worth <u>4</u> visitors (20 ÷ 5). Thursday's total was 252. Wednesday's total was 224. The difference between these numbers is **28**.
56	13 containers	How many containers can be <u>filled</u>? 1500 grams is 1.5 kilograms. Count in 1.5s without going past 20. It can only be done **13 times**. The 14th container wouldn't be full.

TEST 2 ANSWERS >>>

Test 2 English Answers

1	D	The **apostrophe** is placed incorrectly in *haven't*. It should be between the n and t, replacing the letter o from the word *not*. This is known as an apostrophe for contraction.
2	A	The word *let's* needs to have an **apostrophe for contraction**. It is a contraction of two words: *let us*. The apostrophe replaces the letter u. The pair of dashes surround a phrase (*piece by piece*) that is not part of the sentence's natural flow.
3	N	The two words using apostrophes are correct. In the first case, *they'd* is a contraction of *they had*. In the second case, an apostrophe for possession is used: the *identity* belongs to Eric. It is *Eric's*.
4	B	*Believe it or not* has been added to the beginning of the sentence, so **a comma is needed** after the word *not* (it is a dependent clause). Without the comma, the fruit would be actually be called *believe it or not tomatoes*!
5	A	*Capital* **does not get a capital**! In this sentence, it is a common noun (in other sentences it is an adjective: e.g. 'capital city'). Only the names of the cities themselves get capital letters.

6	C	*Carolina, not Bianca,* **was** *planning to be here last night.* The sentence is about *Carolina... not Bianca*! Remove those extra two words to make it more straightforward: *Carolina was planning...*
7	E	**Which** *of these three subjects would you prefer to study next year?* When we have to make a selection from among several items, we use *which*.
8	A	*The gift that was given to my sister and* **me** *shall never be forgotten.* It is a myth that we should always say '__ and I'. Just as with question 6, remove the sister from the sentence; what would we say then? *'The gift that was given to* **me**...
9	C	*As soon as Mr Potter had* **risen**, *he set about his usual morning chores.* The verb *had* is used before the verb we have to choose. This means we are looking for a past participle, not a past tense. Compare these: I **rose**... / I **had risen**...
10	B	*Mine is on display at the Metropolitan Museum, but* **hers** *was removed last year.* Does *mine* have an apostrophe? No. Would *his* have an apostrophe? No. The same rule must apply to *hers*. Possessive pronouns do not get possessive apostrophes.

11	B	The spelling should be **definitely**. It has 'in it' in it! The word *definite* comes from the same root as *finish*. This might help kids remember to spell the word with an F-I-N-I... spelling.
12	N	There are no spelling mistakes. *Restaurant* is a tricky word as it doesn't look the way it sounds. This is because it is a French word. Get kids to sound it out this way when spelling it: *rest... A U (make this sound like 'hey you') rant*.
13	A	The spelling should be **mischievous**. It is connected to the word *chief*, which follows the *i before e except after c* rule. This rule works for words in which the *ie/ei* letter combinations make the long ee sound.
14	D	The spelling should be **conferences**. This is a tricky *-ance / -ence* word. However, if you see the letters *-fer* before the ending, go with *-ence* (e.g. *reference*, *preference*, *circumference*).
15	D	The spelling should be **triangular**. It comes from a noun ending with *-le* (*triangle*). These tend to add *-ar* to their adjective endings (e.g. *circle* becomes *circular*; *single* becomes *singular*; *muscle* becomes *muscular; spectacle becomes spectacular*).

16	B	The passage does not dwell on the horrors of earthquakes (so not option A); it does not say they are preventable (so not option C); it does not tell us how amazing they are (so not option D); and it is obviously not option E!
17	E	See lines 6-10. Earthquakes are caused by the build-up of **pressure** in moving tectonic plates.
18	A	*Intensity* is a **noun**. Words that end with *-ity* tend to be nouns (e.g. *activity*, *sensitivity*, *gravity*, *velocity*). Think of it this way: *intensity* is a *thing*. The adjective would be *intense*. The adverb would be *intensely*. The verb would be *intensify*.
19	D	The term *earthquake-prone* occurs straight after a sentence that says earthquakes *are more common in certain areas* (lines 12-13). This means that earthquake-prone areas **are more likely to experience an earthquake**.
20	C	See lines 32-36. **Fresh fruit** is not mentioned, and would be a pointless thing to put in a kit that has to be stored away for an indefinite period of time!
21	E	Option A is false: earthquakes are more likely to occur in certain areas like the 'Ring of Fire' (lines 11-15). Lines 37-44 show us that option E is correct: **aftershocks of an earthquake can be dangerous**.
22	B	The three actions are mentioned in lines 22-26, as well as lines 52-56. They are **drop, cover, hold on**.

23	pressure	Lines 9-10: *This* **pressure** *is then released suddenly in the form of an earthquake.*
24	debris	*Rubble* and *wreckage* are nouns. This means the word *damaged* cannot meaningfully replace them. Although **debris** (pronounced 'day-bree') may be an unfamiliar word, it can be meaningfully swapped.
25	1960	See line 21: *...the highest number ever recorded was 9.5, which occurred in Chile in* **1960**.
26	9	The prefix *sub-* means 'under'. A subheading is not the same as the title. They are the 'lesser' titles used throughout the text (e.g. *What Causes Earthquakes?*).
27	verb	In each case, these are action words: *'likely to occur...; they can range...; it's important to check...; that can cause....*
28	flashlight	Words such as *non-perishable* and *first-aid* use hyphens, and so are known as hyphenated words, which are not the same as compound words. **Flashlight** (*flash* + *light*) uses no hyphens.

29	E	There are 1000 metres in one kilometre so 2.3 must be multiplied by 1000, making **2300**.
30	A	To move from the top-left **x** to the top-right **x** we go *two squares across and one square down*. To create a parallel line to this, go to the bottom-left **x** and do the same. This will lead to **(5,1)**.
31	B	Barry has saved nine 20p coins (£1.80) so subtract this from £6.00 to get £4.20. How many 20p coins is this? 420p ÷ 20p is the same as 42 ÷ 2. The answer is **21**.
32	A	The bottom two angles in the isosceles are identical. Subtract 38˚ from 180˚ (142˚) then divide by two. Angle A = 71˚. All of the angles in an equilateral are the same (180˚ ÷ 3): 60˚. Finally, subtract 60˚ from 71˚ to get the answer: **11˚**.
33	C	20% is the same as one-fifth. Before the sale, the two items would have cost £20.50 altogether. Find one-fifth of this amount then subtract it from £20.50. 20% of £20.50 is £4.10. The answer is **£16.40**.
34	E	The sequence is +5, +6, +7, +8, +9, +10, which means that the missing numbers are **17 & 38**. If kids don't spot this they can still test out each of the options until one makes sense.
35	D	Volume is found by multiplying the three different measurements, in this case 3.5cm, 9cm and 6cm. It may be best to multiply the 3.5 by 6 first (3.5 x 6 is the same as 7 x 3).
36	C	Multiplying by 1000 then dividing by 10 is the same as multiplying by 100 (i.e. moving each digit two spaces to the left). This means that the two decimal digits will be the in the whole number places. The **7** will be in the hundreds place.
37	B	The entire shape is a square, meaning all four sides measure the same. We can see that the right-hand side will measure 15cm (5 x 3cm). This means that the top of each rectangle is 15cm and the side is 3cm: 15+15+3+3=**36**.
38	D	Subtracting three hours from midnight will bring us to 9pm. Subtracting a further 15 minutes will bring us to 8.45pm. But we have been asked to choose the 24-hour time. Add 12 hours to the hour number: **20:45**.
39	B	Each interval on the jug is worth 250ml. There must be 3250ml in the jug (1000ml = 1 litre). The question asked how many glasses can be filled. We can fill **10 glasses**. This would leave 250ml, which is not enough juice to fill an eleventh glass.
40	E	½ is the same as 0.5. ¼ is the same as 0.25. Add these decimals together to get **0.75**.
41	E	64 is square (8x8) and cube (4x4x4). 100 is square (10x10) but not a cube number. 73 is prime (it can only be divided by itself and one). 75 is not prime (it is divisible by five).
42	C	Harry reads 180 pages from Monday to Friday (36x5). Add Saturday & Sunday's total of 60 to this number: 240. There are 17 pages left to read, so add this as well: **257 pages** in the book.
43	B	Obtuse angles are greater than 90˚ but less than 180˚. We must think of the angle from the 8 to the 12. There are 30˚ between each number on the clock (360˚ ÷ 12). As there are four spaces from 8 to 12, the answer must be **120˚** (30˚x4).
44	C	There are 24 cupcakes altogether. 6 of them are lemon & cinnamon. $\frac{6}{24}$ is the same as **¼**.
45	E	110 is not a square number so we can't know the exact measurement of each side (without a calculator). However, we know that 110 is greater than 10^2 (100) and less than 11^2 (121), so the length of each side is somewhere **between 10cm and 11cm**.
46	C	You can see the vertices in the picture (they are the 'pointy bits' or corners). The shape on the front and back is a hexagon, which has 6 corners. 6 + 6 = **12**.
47	B	No need to perform any multiplications. The digits in the multiplication are in the same order as the ones in the example. This means the answer will have the same digits. 6.46 x 8.5 will be close to 6 x 9. The answer must be close to 54. It is **54.91**
48	B	Test each option out. In option A, the short side is 5cm so the long side would be 10cm (5cm longer). The perimeter would be 30cm. But in option B, if the short side is **4cm**, the long side would be 9cm: the perimeter would be 26cm (4+4+9+9).
49	D	Frank will miss the 0820 bus from Lisburn. His 17-minute walk means he won't be at the stop until 0832. The next Lisburn bus is at 0920, which is **48 minutes** later.
50	E	There are 10mm in 1cm. There are 100cm in 1m. This means there are 1000mm in 1m. 9.6 needs to be multiplied by 1000: **9600**.

51	17	22 kids chose cola. 5 kids chose cherry. 22 - 5 = **17**
52	45	*a* must be 15 (27 - 12). *b* must be 3 (three cubed, or 3x3x3, is 27). 15 x 3 = **45**.
53	B	Travel southeast from D. This means cutting perfectly through the opposite corner of each square. We will not go through C as we do this. Only the letter **B** is perfectly southeast of D.
54	2.72 kg	40% is four-tenths, or two-fifths, so either divide 6800g by 10 then multiply by 4, or divide it by 5 then multiply it by 2. It will make 2720g, which must be converted into kilograms: **2.72kg**.
55	4	Hannah has rolled a 10. Her next roll will be 1 at the least or 6 at the most, so her final total will be between 11 and 16. The only multiple of 7 in this range is 14, so she rolled a **4**.
56	66 cm^3	The original cuboid's three measurements were 8cm (width), 6cm (height) and 2cm (depth), making a volume of $96cm^3$. The cut-out space measures 5cm by 3cm by 2cm ($30cm^3$). $96cm^3$ - $30cm^3$ = **$66cm^3$**.

TEST 3 ANSWERS >>>

Test 3 English Answers

1	B	A **question mark** is needed instead of a comma: '*Can you believe this?*'
2	A	'*When things calm down...*' There **should not be an apostrophe** in this word. It is not short for 'thing is' or 'thing has'. The *things* do not possess anything either. It is just a simple plural.
3	B	**There should not be a question mark**. Toby is making a remark about his day, not asking a question. An exclamation mark could be used, depending on the tone of voice that Toby uses. If not, then a comma will suffice.
4	A	The word should be *children's*. **The possessive apostrophe** goes before the letter s. Possessive apostrophes are written immediately after the name of the owner: in this case, *children*.
5	D	While semi-colons do not have to be understood in any depth, one thing must be known about them: they can't be placed at the very end of sentences. A **full stop** is needed instead.

6	A	*One thing I know: it* **could** *be next year by the time we discover the true culprit....* No other option works (*could've* would have to be followed by *been*).
7	D	*It was too quiet. There were far* **fewer** *fans at the match than the club had expected.* When you can count the actual number, use *fewer* (one fan, two fans, fewer fans). When you can't give a number, use less (~~one water~~, ~~two waters~~, less water).
8	C	*The Jessop family just made it in time* **for** *their flight to Paris.* The preposition *for* is commonly used after the word *time* (e.g. 'time for dinner', 'time for a change').
9	E	*Yasmin was a very intelligent girl but, as she easily became nervous, she rarely did* **well** *in her school tests....*' An adverb is needed to describe the verb *did*. We also know that something went wrong for Yasmin so she rarely did *well*.
10	A	'*Neither the pupils* **nor** *the parents were aware that school would be closed on Monday...*' The pairs of words that go together are *either / or* (positive) and *neither / nor* (negative).

11	D	The spelling should be **business**. Although it technically doesn't come from the word *busy*, this is still a useful way to remember its spelling pattern (spell *busy*, exchange the y for an i, then attach the *-ness* suffix).
12	D	The spelling should be **whole**, which is a hompophone of *hole*. *Disastrous* has been spelled correctly (the letter e at the end of *disaster* disappears when the adjective is created).
13	C	The spelling should be **unsuccessful**. There must be a double c in *success*. This is because the first c makes a 'hard' k sound and the second c makes a 'soft' s sound (because it is followed by an e). These combine to make an x sound.
14	N	There are no mistakes. *Curiosity* is an unusual spelling because the adjective *curious* has the letter u before the s. But when we say *curiosity* we emphasise the *'os'* syllable strongest.
15	B	The spelling should be **environment**. The letter n in the middle is often missed. This is because we don't really sound it out when saying the word. There is *iron* in the *environment*.

16	C	Ape is **confused**. This is the best alternative to *indecisive*. He cannot decide which thing to do. He is not necessarily *lazy* (option D) as some of his options are active (e.g. line 5). He only becomes *determined* (option B) towards the middle of the poem.
17	D	Lines 1 & 2 in each verse rhyme with each other (e.g. ape/escape); lines 3 & 4 do the same (e.g. do/clue).
18	C	Restless King was advised to **make definite decisions**: *Get up! DO SOMETHING!* (line 48).
19	D	Look at lines 37-40. No **elephants** are mentioned.
20	B	The jungle doesn't have a literal gate, even though the phrase makes it sound like it does. This is a **metaphor** – when we know that the poet/author doesn't literally mean the thing he just said (e.g. life is a rollercoaster).
21	A	The poem is meant to be **playful** (even if kids don't find it to be much fun!). The rhythm of the poem and the far-fetched nature of the story should lead the reader to think of it as light-hearted.
22	D	*Unknown* is a **noun** in this line: *The great unknown beyond his door*. Although the word is usually an adjective (e.g. 'an unknown person sent me a gift'), in this line it is a noun because it is being used as the name of a place.

23	aspire	See line 26: *He grew to weary to* **aspire** [too tired to hope, too tired to dream].
24	pros and cons	Kids should look for a three-word phrase (the second word is likely to be *and*). So it's either **pros and cons** (line 14) or *head and screw* (line 13)... which is not a phrase!
25	wherever	The apostrophe has replaced the letter v. **Wherever** *he went, he now could choose...*
26	12	*Or lay around just like a loaf*. Similes compare one thing with something different because of one thing they have in common. They use the words like or as. Children must write the line number.
27	seek	**Seek** / *sought* may be the most difficult present / past tense to remember. It's worth reminding kids about this pair of words on a regular basis.The present tense form is on line 30.
28	34	The new name suits Ape's new character: *Decisive Ape*. Again, the answer is the line number, not the name. The name appears again on lines 39 and 49 but children must find its first mention.

29	D	£12.50 x 12 (months) = **£150** This multiplication can be simplified: double £12.50 to make £25, halve 12 to make 6, then multiply 25 by 6.
30	E	These are *triangular numbers*: 1 *(+2)* 3 *(+3)* 6 *(+4)* 10 *(+5)* 15, etc. Kids could write out a careful list to get the 7th number: **28**. Or they could use the 'trick' 7x8 then ÷2. If they had to find the 12th number (as another example) it would be 12x13 then ÷2.
31	A	A triangular-based pyramid is *made up entirely of triangles*. The top and bottom triangles on Net C would fold in on each other, so it must be **Net A**.
32	B	**0.64** is the same as 64 hundredths. Kids should visualise the '64' finishing in the hundredths place.
33	B	All sides measure the same on an equilateral triangle, so if the perimeter is 12cm, one side measures 4cm. The quadrilateral (or rhombus) consists of eight sides from these triangles. 8 x 4cm = **32cm**.
34	C	Draw a timeline then subtract two hours from 16:15 (14:15), 15 minutes (14:00), and 20 minutes (13:40). Convert this to 12-hour time. The answer is **1.40pm**.
35	A	By reading the scale of the map we can see that the kilometres value will be four times greater than the centimetres value. We have been told the kilometres measurement (7.2) so this must be divided by 4 to get the centimetres measurement: **1.8cm**.
36	C	The letter **C** does not reflect a corner of the quadrilateral. It is just one square away from the mirror line. Count one square to the left of the mirror line and you will not find a corner of the quadrilateral.
37	D	Good estimates would be £1.45 times 6 or times 7. £1.45 x 6 = £8.70. Another £1.45 would take Tommy past his spending limit of £10. His change will be **£1.30** (£10 - £8.70).
38	B	Rhombuses have two identical acute angles, and two identical obtuse angles. One of each angle will add to make 180˚. Subtract 64˚ from 180˚ to get **116˚**. The same rule applies to parallelograms.
39	C	The tally marks add to make 40 altogether. 10 children chose Dobble, which is one-quarter, or **25%**, of 40.
40	C	8 is a factor of 24. 9 is an odd number. **10** is neither prime, odd, nor a factor of 24, so it is the smallest of the options that can't be placed into any circle in the Venn Diagram.
41	B	There are twelve edges on a cuboid. Whenever we see three different measurements, we know that there are four of each of these edges. 15cm + 7cm + 4.5cm = 26.5cm. 26.5cm x 4 = **106cm**
42	E	Round each number in the calculation to its nearest ten. It becomes 90 x 90 (or 9 x 9 then add the two zeros), which makes **8100**.
43	B	The rectangle's area is 30cm² (5cm x 6cm). The base of the triangle is 5cm. The height of the triangle is 2cm (8cm - 2cm). This means its area is 5cm² (5cm x 2cm then ÷2). Together, the area of the trapezium must be **35cm²**.
44	A	Find the length of time between 9.15am and 10.40am (1 hour 25 minutes...or... 85 minutes), then subtract 15 minutes for the break. This makes 70 minutes. The journey home takes 45 minutes, making **115 minutes** on the bus altogether.
45	D	Test each option out... or... as there is the same number of each coin, we can pair each 50p with a 20p. These combine to form a 70p pair. How many times does 70p fit into £6.30? This is the same as 63 ÷ 7. The answer is **nine**.
46	D	Add the eight scores together to get 64. Divide this by eight (the number of numbers) to get the mean: **8**.
47	A	The first number in the calculation has been tripled: it was an 8, but now is 24. The important fact is that it is being divided by the same number. This means that the answer also has to become three times greater: 0.25 x 3 = **0.75**.
48	B	135˚ is the same as three compass points. Andrew moves anticlockwise from northeast through the following points: north, northwest, then finally **west**.
49	C	We are finding the percentage of girls, not boys. There are 18 girls (45 - 27). Simplify this fraction down to two-fifths. This is equal to four-tenths, or **40%**.
50	E	Sergio needs to buy five cartons of milk (four would only be 2400ml – or 2.4 litres – which is not enough for the recipe). So he has 3 litres of milk (3000ml) and uses 2.5 litres (2500ml) in the recipe, meaning that the final carton still has **500ml** in it.

51	8	Even though the measurement of each side is doubled, the volume of the box is eight times greater than the volume of the small cube. It has been doubled in length, doubled in height, doubled in depth.
52	373	Two hundreds = 200. 17 tens = 170. 30 tenths = 3 (or 3.0). 200 + 170 + 3 = **373**
53	11.5 cm	All sides measure the same on an equilateral (see question 33). Divide the perimeter by 3. 34.5cm ÷ 3 = **11.5cm**
54	Thursday	From Tuesday 31st October, there are 23 days until 23rd November. 23 days is three weeks and two days. Three weeks will bring us to another Tuesday. Add two days to Tuesday to reach **Thursday**.
55	5.76 kg	720g x 8 = 5760g. Divide this by 1000 to get the number of kilograms.
56	(0,6)	Look at (6,6). If we can move six squares to the right to plot another corner, then we can also move six squares to the left. This would create a parallelogram that 'leans' to the left.

TEST 4 ANSWERS >>>

Test 4 English Answers

1	N	There are no mistakes. The sentence that Jake wrote needs an exclamation mark, but not the sentence that you are reading!
2	C	The *topic* belongs to *October*. This means that *October's* should have a **possessive apostrophe**. As *Ancient Greece* is a recognised historical name, both words must be capitalised (not just *Greece*).
3	C	It should be *many moons ago*. This is a simple plural so **doesn't need an apostrophe**. It is not short for two words (e.g 'moon is') and the moons do not possess anything in the sentence. The comma that separates *many* from *many* is correct.
4	B	Closing speech marks always need a punctuation mark before them. In this case, a **comma** is required. It should be *'If you finish your homework,' Dad informed Jess...* The possessive apostrophe in *Abby's* implies *Abby's house*.
5	B	Even though the exams will not rigorously test a child's understanding of semi-colons, they should still be able to spot when they're out of place. This is a list of single words and so it should be a **comma** after *raisins*, not a semi-colon.

6	B	*'Ask* **yourself** *if there are any other ways to solve the problem.'* For option A, the word should be *me*. For option C, the word should be *him*. For option D, the word should be *her*. For option E, the word should be *them*.
7	E	*While cleaning the attic, I came* **across** *a box of old photographs from my childhood.* To *come across* something is to find or discover it.
8	A	*The elephants passed by the lake and* **drank** *from it.* The past tense of *drink* is *drank*. In this sentence, we need a simple past tense. The verb is not preceded by a word like *had* or *have* so we should not use the past participle *drunk*.
9	C	**Whenever** *you arrive, just ring the doorbell, and I'll let you in.* Rearrange the sentence order: *I'll let you in... when you arrive*. Our choice of word must be connected to the word *when*.
10	E	*'I remember it* **clearly**,*' said Mum, 'as if it were yesterday.'* An adverb is needed to describe the verb *remember* (*clear* is an adjective). It can only be options D or E but, as *probably* doesn't make sense, it must be *clearly*.

11	A	The spelling should be **interestingly**. The first e in this spelling is often missed. Teach kids that there is an '*e resting*' in the word *interesting*.
12	B	The spelling should be **calendar**. *February* is spelled correctly. Just as there are two months in the calendar that begin with A (April & August), so there are two letter As in the spelling of *calendar*.
13	D	The spelling should be **guidance**. There is no easy way to decide between *-ance* and *-ence* in words like this. It's best to come up with a silly but memorable story for this word (e.g. I want to draw a *guy dance*; could you give me some *guidance*?).
14	A	The spelling should be **discipline**. There is an unexpected c in this word. It sounds like an s because the letter i is after it (consider how this letter combination sounds in words like *circle* and *racing*).
15	A	The spelling should be **intelligently**. The vowels in *intelligent* follow an *i-e-i-e* pattern.

16	B	See lines 3-4: **Walter** is in *Rainbow Valley*.
17	A	The majority of this passage, and the focus of the women's conversation, is about **church matters** – particularly the new minister. This subject first gets brought up in line 19, and continues through to the end of passage.
18	E	**Miss Cornelia** is saying these words. It was she who spoke in lines 13-15. Anne replies in lines 16-17. A new paragraph is taken in line 18 (therefore a new speaker). No name is given so we must assume it's back to Miss Cornelia. Susan rejoins in line 21.
19	B	Look at lines 24-26. First, Miss Cornelia is not impressed by the focus on Mr Meredith's looks. She then complains about him having no common sense. After this, **she softens her tone by saying something positive about him**.
20	D	See line 23: Anne asks if Mr Meredith is *handsome... her eyes* bright. She is primarily interested in his looks. Susan and Miss Cornelia respond with a sigh and a groan. Miss Cornelia directs the conversation to his personality and skills.
21	C	See lines 29-30. Mr Meredith was likely not called to a town church because he was *so moony and* **absent-minded**.
22	D	The simile is on **line 38**: *he was like a chip in porridge*. Similes are phrases that compare two things by using the words *as* or *like*.

23	85	See line 16: *'Marilla is* **eighty-five**,*' said Anne with a sigh.*
24	Cuthbert	The answer is on line 13. By following the conversation from lines 11 to 17, we can tell that Marilla is **Miss Cuthbert**.
25	can do no wrong	This phrase is on line 12. If someone **can do no wrong**, they must be 'perfect'.
26	busy & idle	**Idle** means 'lazy, motionless, doing little or nothing'... the opposite of **busy**.
27	4	The **four** candidates were *Mr Meredith*, *Mr Folsom*, *Mr Rogers* and *Mr Stewart*. We are not told if *Caleb Ramsay* was even in the church: it was his sheep that had wandered in!
28	adverb	*Hardly* describes the verb *wait*. *Terribly* describes the verb *spoil*. *Simply* describes the adjective *wonderful*. *Loudly* describes the verb *bleated*. **Adverbs** describe both verbs and adjectives.

Test 4 Maths Answers

29	D	The correct calculations are **x7 then -6**. The sequence has to work for both pairs of numbers in the function machine. Options A & C only work for the first pair of numbers (8 becoming 50). Options B & E only work for the second pair (12 becoming 78).
30	D	Subtracting 100 minutes is the same as subtracting 1 hour and 40 minutes... which is the same as subtracting 2 hours and adding 20 minutes back on again. So the time is 17:40, which is **5.40pm** as a 12-hour clock time.
31	C	Parallel lines run in exactly the same direction and would never meet, even if they were extended. **Line C** is parallel to Line F.
32	D	Three-eighths of 80km is 30km, so, after the first hour, the businessman has 50km remaining. Three-fifths of this 50km is 30km again. This means he travelled 30km in hour one, and 30km in hour two, making **60km** in total.
33	E	The total of five items was divided by five to find the mean. □ ÷ 5 = £20. The total must have been £100. This means all five numbers add to make £100. The first four add to make £72. The fifth price must be **£28** (£100 - £72).
34	A	A parallelogram like this has **zero** lines of symmetry. A square has four; a rhombus is a 'leaning square' and has two fewer. A rectangle has two; a parallelogram is generally considered as a 'leaning rectangle' and has two fewer.
35	E	20% is one-fifth. Make equivalent fractions to discover which one doesn't work. **One-fifth can't be the same as two-fifths**.
36	B	The fewest coins to make £3.97 is **seven**: £2, £1, 50p, 20p, 20p, 5p, 2p.
37	E	A square-based pyramid has 8 edges. A sphere has 0 edges. A triangular prism has 9 edges. A cuboid has 12 edges. **A hexagonal prism has 18 edges**.
38	B	The sequence is -9, -8, -7, -6, -5. The missing numbers are **59 & 48**.
39	B	The line graph shows when the children were in Carrickfergus. It is the horizontal line at the top. This 'flat' line starts at 1130 and finishes at 1515. This means they were there for **3 hours and 45 minutes**.
40	D	The intervals on each ruler are different. The intervals on the top ruler are 1mm each. The intervals on the second ruler are 2mm each. The first line is 12.5cm. The second line is 9.8cm. This makes 22.3cm altogether, which is the same as **223mm**.
41	B	Rectangle A has an area of $42cm^2$ (10.5 x 4). Rectangle B has an area of $30cm^2$ (12 x 2.5). The difference is **$12cm^2$**.
42	A	There are 420 pieces still in the box (1000 - 580). As a fraction of the entire jigsaw set, this is $^{420}/_{1000}$, which can be simplified to make $^{42}/_{100}$. If we have a fraction that has 100 as a denominator, we can see the percentage in the numerator: **42%**.
43	C	Six pancakes is six-tenths of ten pancakes, so divide the amount of milk (300ml) by ten (30ml) then multiply by six to get **180ml**.
44	C	The smallest number is 4.375. It has fewer tenths than 4.4. It has fewer hundredths than 4.39. 4.375 x 100 = 437.5... 437.5 subtract 4 is **433.5**.
45	E	There are 60 children in total. 6 chose table-tennis, which is one-tenth of the children. There are 360˚ in a pie chart. One-tenth of this is **36˚**.
46	A	Be careful to not count the one-digit square numbers (1, 4 and 9). The two-digit square numbers are 16 (4x4), 25 (5x5), 36 (6x6), 49 (7x7), 64 (8x8) and 81 (9x9). The next square number (100) is a three-digit number so there are **six** in total.
47	E	Volume is found by multiplying the three measurements together: 8cm x 3cm x 1.5cm. This makes **$36cm^3$**.
48	A	Work backwards through the story, just as you would often do with a function machine. 1408 divided by two (the opposite of 'double') is 704. Then subtract 300 (the opposite of add 300) to get **404**.
49	B	One wristband costs 90p (£3.60 ÷ 4). One rubber ball costs 70p (£2.80 ÷ 4). Andrea buys six wristbands (90p x 6), which is £5.40. She buys three rubber balls (70p x 3), which is £2.10. These add to make **£7.50**.
50	D	Keith ran 400m (or 0.4km) plus 6.9km to get to Newtownards. He has to run exactly the same distance to get back to his car. 6.9 + 0.4 = 7.3. Double this to get **14.6km**.

51	18 kg	The difference between the two weights is the weight of the rucksack so subtract 35.2 from 53.2.
52	124	The first five cube numbers are 1 (1x1x1), 8 (2x2x2), 27 (3x3x3), 64 (4x4x4) and 125 (5x5x5). The range is the difference between the smallest and largest numbers. 125 - 1 = **124**
53	60˚	The two known angles make 200˚ so A and B must make 160˚ (360 - 200). The difference between A and B is 40˚. Subtract this 40 from 160 (120) then halve this number to finder the smaller angle: **60˚**.
54	126 cm	If the perimeter of a regular hexagon is 42cm then each side must measure 7cm (42 ÷ 6). There are 18 sides around the pattern. 18 x 7 = **126**
55	32˚C	The difference between 25˚C and -7˚C is found by adding 25 and 7. Do not accept -32˚C as an answer.
56	13.5 seconds	10% (one-tenth) of 15 seconds is 1.5 seconds. However, as Grace is faster than Tori, it takes her less time to run the race. Subtract 1.5 from 15.

TEST 5 ANSWERS >>>

Test 5 English Answers

1	D	**There is no full stop at the end of the sentence**. It should be placed before the closing speech mark.
2	A	A **possessive apostrophe** is needed for the word *brother's*. *Emily's brother* has a *book – in the same way that Mya's sister* has a *schoolbag*. It is therefore possible to have two consecutive words with possessive apostrophes.
3	C	If *Venus* and *Mercury* must be **capitalised**, then so must *Earth*.
4	N	There are no mistakes. The opening clause (*Tilting his head to the right*) must be separated from the main sentence by a comma. The comma is also appropriate between the two adjectives that describe the *inscription*.
5	A	A **question mark** is needed for Uncle Bill's first sentence. *Someone's* is a contraction of 'someone is'. We have to assume that he used some emotion in his second sentence, which would make the exclamation mark appropriate.

6	B	*As we walked slowly* **past** *the toy store, we came to Mr Somerville's old repair shop*. The toy store is not the destination in this sentence, but must be walked past in order to reach the repair shop.
7	D	*The syrup-covered pancakes* **were making** *me drool like a dog at the foot of the* dinner *table.* It can't be *were causing* because we would need to add the word *to*: *...were causing me to drool...*
8	C	*Gertrude had never in her long, miserable life* **given** *anything to anyone.* We must use a past participle. The word *had* is at the start of the sentence and is directly connected to our answer. Instead of *gave*, we must say *had given* (I **gave** up / I **had given** up).
9	A	*We call him 'Chatty Charlie' because he is* **seldom** *quiet.* Even if this is an unfamiliar word, kids should know to eliminate the other options. *Charlie* is *chatty*, which means he is not *quiet*. All the other options would suggest the opposite.
10	E	*'Whatever excuse you give me,' Mrs Patterson warned Jordan,* **'it'd** *better be a good one!'* Although it sounds strange when said by itself, *it'd* is a contraction of *it had* (or 'it would' on other occasions).

11	D	The spelling should be **disappointing**. It has a double p rather than a double s. *Dis-* is a common prefix (*disappear*, *dislike*, *discourage*). In this sentence, it is added to the word *appointing*, which has a double p.
12	N	There are no mistakes. *Difficult* contains a double f. It is sometimes mistakenly spelled as 'diffacult'.
13	D	The spelling should be **growl**. It has a similar ending to *prowl* and *scowl* (and *owl*!).
14	B	The spelling should be **exercises**. No excuses for missing this one: the word is written at the top of the page (and the one before, and the one before)!
15	D	The spelling should be **foreign**. The *i before e except after c* rule does not apply to words in which the two letters don't make the long ee sound. Try this sentence to memorise the spelling: *Funny octopus rubs eggs into George's nose*.

16	B	The prefix *tri-* means 'three' (e.g. *tricycle*, *triangle*). The answer is **trilogy**.
17	D	See line 4: *These movies, directed by* **George Lucas**...
18	A	The overall purpose of the report is to inform us of the history of the Star Wars movies – their storylines, their success, how they influenced culture, etc. It was **a significant movie franchise**.
19	B	Although Darth Vader is as notorious as a villain, it is actually Han Solo's description that paints him as a lawbreaker: **a rebellious smuggler**. Darth Vader's description as *a ruthless enforcer* does not suggest that he is a criminal – just powerful.
20	C	*Twist* is a **noun** in this sentence. If a story has a twist, it has something. Nouns are things! Just because *twist* is often used as a verb doesn't mean it always has to be one.
21	E	Options B and D are incorrect because they use the word *entirely*. The Force is neither completely good nor completely evil. But it can **improve speed and agility, among other things** (see lines 32-33).
22	D	The prefix *pre-* means 'before' (e.g. *preview*, *preschool*). A **prequel** is a story that occurs before another one.

23	1977	Lines 41-42: *'Star Wars: A New Hope' was released in* **1977**...
24	$538 million	See lines 43-45: *'Star Wars: The Empire Strikes Back' and 'Star Wars: Return of the Jedi' were also major box office hits, earning* **$538 million** *and $475 million respectively.*
25	Death Star	The answer is on line 11. *Death Star II* is not mentioned in lines 6-13 so do not accept this as an answer.
26	What Happened Next	Lines 50-51 contain the connection: *a vast array of merchandise* means the same as *a wide range of goods and products.*
27	hangs in the balance	See line 25: *On a knife-edge* is a metaphor worth learning, particularly when describing tense moments in stories. It suggests that events could 'go either way'.
28	adjective	*Groundbreaking* describes the noun *effects*. *Memorable* describes the noun *twist*. *Original* describes the noun *trilogy*. *Vast* describes the noun *array*. If a word descibes a noun, it is an **adjective**.

29	E	64 must be multiplied by 8 to find the answer: **512**.
30	A	Add the four players' amounts together (£7397) then subtract it from the total to get Maggie's amount: **£1500**.
31	B	The top 'half' of the shape is not actually half! It is four squares high, whereas the bottom 'half' is five squares high. The shape is a kite, which has **one vertical line of symmetry** down the middle (i.e. it can only be folded across the way).
32	C	The fraction $\frac{25}{40}$ can be simplified to make $\frac{5}{8}$, (divide both the numberator and denominator by five).
33	C	Options A and B both make £9. Option D makes £9.75. Option E makes £8. **Option C** makes £10 (sixteen 50p coins = £8; twenty 10p coins is £2).
34	D	The total number of degrees goes up by 180 each time an extra side is added to the shape. So, if a six-sided shape has 720˚ in total, then we need to add 180˚ twice to get the total number of degrees in an octagon (an eight-sided shape).
35	E	The fastest way to find perimeters of shapes like these is to pretend they are full rectangles (in other words, no corners have been removed). They will have the same perimeter. This one would be 9cm by 7cm, which makes a perimeter of **32cm**.
36	D	For week 1, *Top Table* had three different scores in three different categories. Read across from *Week 1* on the table. They scored 5 in *General Knowledge*, 9 in *Music & Film*, and 6 in *Sport*. This makes a final score of **20 points** (5 + 9 + 6).
37	A	3% means 3 out of 100 - or three hundredths, so the digit 3 must be in the hundredths place: **0.03**.
38	D	Study the length of the brown line. Left to right, you travel seven squares across then four squares up. Then go back and imagine (4,4) is connected to **(11,0)**. It would be seven across and four down: the same length! It can't be scalene.
39	E	A prime number has exactly two factors: 1 and itself. There are **four** prime numbers between 10 and 20. They are 11, 13, 17 and 19.
40	C	The intervals are worth 200ml each (five steps to get from one litre to the next), so we are 200ml below 3 litres: **2800ml**.
41	A	The McClean family pays for three nights. £120 x 3 = £360. They get 10% off. £360 - £36 = £324. The Turbotts pay for five nights. £120 x 5 = £600. They get 20% (one-fifth) off. £600 - £120 = £480. The difference between these totals is **£156**.
42	E	A multiple of 3 can be easily spotted by adding its digits together; their sum will also be a multiple of 3. There are four multiples of 3 on the cards: 42, 27, 51 and 39. There are only two prime numbers: 19 and 47.
43	B	A cylinder has no vertices. **A cube has eight vertices**. A cone has one vertex. A square-based pyramid has five vertices. A triangular prism has six vertices.
44	B	160 minutes is 2 hours 40 minutes. This must be subtracted from 19:05. Subtract 2 hours, then 5 minutes, then 35 minutes to get to 16:25. This is the same as **4.25pm**.
45	A	Each base angle on the isosceles is 65˚ (180˚ - 50˚ then ÷ 2). Angles next to each other on a rhombus add to make 180˚ so the bottom-left angle is 60˚. Angle A is stuck between a 65˚ and a 60˚ angle. Together, all three (on a straight line) must make 180˚.
46	D	To find the volume of a cube (or cuboid), multiply length by breadth by height. In the case of a cube, each of these three measurements is the same. 2cm x 2cm x 2cm = **8cm³**
47	C	The range is the difference between the smallest and largest values in a list. The shortest measurement is 34.7cm. The longest measurement is 36.2cm. This is a difference of 1.5cm, which is the same as **15mm**.
48	C	One hundredth is 0.01. 5100 subtract 0.01 is **5099.99**.
49	D	50cm fits into 4m eight times, so 8 tiles will fit across the wall. 50cm fits into 2.5m five times, so there will be 5 rows of tiles. Multiply 8 by 5 to get the final answer: **40 tiles**.
50	B	Go in reverse: clockwise. 225˚ is a half-turn plus one more point around the compass (180˚+45˚). This will bring us from southwest to **east**.

51	1848	Compare the two calculations. The first number in the new calculation is 10 times smaller. The second number is 100 times greater. This means the answer will be 10 times greater (÷10, x100).
52	56 m	0.07km is 70 metres. (0.7km would be 700m; 0.007km would be 7m). 70m subtract 14m is **56m**.
53	44	14¾ is the same as 14.75. Multiply this by 3 to get 44.25 (or 44¼). Subtract the quarter to get **44**.
54	Friday	There were 20 visitors on Thursday. 50% of this is 10 visitors. Add this 10 to get 30, which is the number of visitors on **Friday** (the increase from Wednesday to Thursday was 100%).
55	8 cm	If the perimeter is 20cm, then one of each measurement will add to make 10cm. Two numbers add to make 10, with one being six more than the other. These numbers would therefore have to be **8** and 2.
56	729	Work from right to left and perform inverse calculations. 67 plus 14 is 81. 81 multiplied by 9 is **729**.

TEST 6 ANSWERS >>>

Test 6 English Answers

1	B	*Thursday* is a proper noun so needs a **capital letter**. The pair of dashes correctly separate the extra information from the main clause of the sentence: *The trip was postponed due to bad weather.*
2	A	The *lake* belongs to the *forest*. This means that it should be *the forest's lake*. A **possessive apostrophe** is needed.
3	C	Closing speech marks always need a punctuation mark before them, not after. It should be *'When I call out your name, Sam,'*... The comma should be before the closing speech mark.
4	D	The second sentence is also a question, so needs a **question mark**. Not only this, we never use full stops at the end of direct speech if the sentence is continued by the narrator.
5	N	There are no mistakes in this sentence. The comma is appropriate after the word *news* as it separates the introductory clause from the main clause.

6	B	*As soon as Emma had entered the living room, the surprise guests* **sprang** *out*. The simple past tense for *spring* is needed, not the past participle. Compare these: the guests **sprang** out / the guests **had sprung** out.
7	E	*'I've already* **said** *this a thousand times and won't tell you again!'* Because the word *have* is already present within the word *I've*, we can't choose an option that contains *had*/*have*.
8	B	*It is considered to be* **an** *honour to meet members of the royal family.* The h at the beginning of *honour* is silent, so we treat the word as if it begins with a vowel, which means we precede it with *an*.
9	A	*Those going through tough times should never to suffer in* **silence**. A noun is needed, so we must use the word *silence*.
10	C	*Without warning, the crowd spilled* **onto** *the streets, cheering loudly for their team*. Something 'spilling' will go *onto* something (or *over* it - although this would sound less appropriate when used for people).

11	A	The spelling should be **heights**. The e is often missed. As *height* is often given as a number/measurement, show children that we can read the number *eight* when spelling *height*.
12	C	The spelling should be **independence**. Emphasise the *pen* and *dent* at the end of *independent* (like a *pen* with a *dent* in it). This could remind kids to choose an *-ent* / *-ence* ending.
13	N	There are no spelling mistakes There is no double t in *edited* / *editing* (or *editor*). This is because we don't stress the second syllable in *edit* (*edit*). The emphasis remains on the first syllable so doubling the t would mistakenly shift the emphasis.
14	A	The spelling should be **gadget**. A *-dj-* spelling is highly unlikely (although it does exist in words like *adjective* and *adjust*). The *-dg-* spelling is much more common (e.g. *badge*, *fridge*, *hedge*).
15	D	The spelling should be **deceived**. This is a perfect example of the *i before e except after c* rule. The rule only counts for words in which the two letters make the long ee sound.

16	C	The poem describes a sailor's love for being out at sea. He feels '**at one**' with it. It is his (metaphorical) home.
17	A	Line 3 tells us that the sea is **without boundary** *(without a bound).* Line 5 tells us that it can be **wild** (the sea rises so high that it *plays with the clouds*). Line 6 tells us that it can also be calm (like a child in a cradle).
18	E	The sea cannot literally mock something or play with anything. These are things that people do, which means that the poet is using **personification**. He is giving the sea a personality.
19	D	Just as any infant or small creature wants to be protected by its mother, the poet felt that the sea was his *nest,* where a bird would return to its mother. It was his 'home'; **it was like returning to his family**.
20	C	The sea is not the poet's actual mother! He is using **metaphor**. This is when we say one thing is another thing (even though it isn't) because they share something in common.
21	E	The poet is describing his first day at sea as if it were the day of his birth. He found **a new purpose in life**. Again, this is a metaphor.
22	E	*Struggle* means the same as **strife**. *Aloft* is similar in meaning to 'up high'; *tempests* are storms; a *porpoise* is a sea creature.

23	2	*Ocean's* (line 30) and sailor's (line 32) are the only uses of a possessive apostrophe in these verses. *I've* (line 31) uses an apostrophe for contraction (*I have*).
24	Death	A proper noun is a name for a particular person, place or thing and begins with a capital letter (e.g. *Frank*, *London*, *World Cup*). In this case, the poet has turned **Death** into an actual person.
25	(every wild) wave	Read lines 15 & 16 together. **Every wild wave** hides the moon and whistles a tune.
26	With wealth to spend and power to range	See line 33. **Wealth** refers to the poet's *money*. **A power to range** is an ability (or *freedom*) to go wherever he wishes.
27	the blue above	**The blue above** (line 9) is another way of saying *sky*. *The blue below* must be the sea.
28	noun	All of these words, within the poem, are things. They must therefore be **nouns**. Note the words that come before these words (*a*, *the*, *in*): these precede nouns.

Test 6 Maths Answers

29	C	We must work out how many times 24 fits into 480. It fits twice into 48, which is ten times smaller than 480, so multiply two by ten to get **20**.
30	B	Moving clockwise, southeast is the very next point on the compass after east. The movement will be half of a right angle (90˚), so must be **45˚**.
31	A	Make every value either a decimal or a percentage. ⅘ **is 80%**, 0.7 is 70%, ⅔ is 66⅔%, 0.68 is 68%.
32	D	The pictures show the first three triangular numbers: 1 *(+2)*, 3 *(+3)*, 6. To find the fourth triangular number we add four to get 10. But this must be multiplied by 3 (there are three matchsticks per triangle): **30**.
33	A	Write the three measurements onto the pile of boxes. It will be 2 metres wide, 1 metre deep, and 1 metre high. 2m x 1m x 1m = **2m³**.
34	B	A digit that is two places after the decimal point is in the **hundredths** place.
35	B	Each rising side on the isosceles triangle is 4cm. This means that the side below both 54˚ angles is 5cm (13 - 8). Now look at the pentagon. Its outside is made up entirely of these 5cm sides. We can tell this from the angles. The perimeter must be **25cm**.
36	C	22:45 is 10.45pm. It takes 15 minutes to reach 11pm, then another hour to reach midnight (12am). 1 hour and 15 minutes is the same as **75 minutes** (60 + 15).
37	C	Figure A is 144cm. Figure B is 126cm. To find the measurement that is halfway between these two, add them together then divide by two (the same method as finding the mean). 144 + 126 = 270... 270 ÷ 2 = **135**.
38	A	The Ham & Pineapple section must be 45˚ (135˚+90˚+90˚ then subtract the total from 360˚). This is one-eighth of the pie chart. There are 32 pupils, one-eighth of which is **4**.
39	B	Kids should not perform the actual calculation! They can estimate. Round 8.9 up to 9. Round 201 down to 200. 9 x 200 = **1800** *(the actual answer is 1788.9)*
40	D	Perpendicular lines meet at a right angle (such as two sides that meet on a square). Line BD is vertical. We must therefore look for a horizontal line. The only horizontal line among the options is **line CF**.
41	B	40% is the same as ⅖ or ⁴⁄₁₀, so divide 250 by 5 then multiply by 2 (or divide 250 by 10 then multiply by 4). 250 ÷ 5 = 50... 50 x 2 = **100**
42	C	1 is not a multiple of 4. Neither is 2. The smallest number that suits all three conditions is **4**. It is a factor of 32 (4x8). It is a square number (the answer to 2x2). It is a multiple of 4 (4x1).
43	D	When folded into a cube, the three dimensions will each be 4cm. 4cm x 4cm x 4cm = **64cm³**
44	E	It may help to visualise the 0.6 as a 60, and the 0.7 as a 70. The steps between these numbers would be easier: 62, 64, 66 and 68. The arrow is pointing at where the 64 would be. But, as it is a decimal number, it is really pointing at **0.64**.
45	B	All sides measure the same on a square. Its entire area must be 64cm² (8cm x 8cm), The coloured part of the square is one-eighth of the entire square, meaning that its area is **8cm²**.
46	D	The first five cube numbers are 1 (1x1x1), 8 (2x2x2), 27 (3x3x3), 64 (4x4x4) and 125 (5x5x5). Add these together then divide by five to get the mean. They add to make 225. 225 ÷ 5 = **45**.
47	C	A cuboid has 12 edges (even). A triangular-based pyramid has 6 edges (even). **A triangular prism has 9 edges (odd)**. A hexagonal prism has 18 edges (even). A square-based pyramid has 8 edges (even).
48	E	Multiples of three have digits that add to make another multiple of three. The answer is **222** (2+2+2=6, which is a multiple of 3).
49	D	¹⁸⁄₆₀ can be simplified to make ³⁄₁₀. Divide both numerator and denominator by six.
50	B	Work backwards through the problem. Bus B finished with 30 passengers. During its journey, 5 got on and 12 got off. Reverse this: add 12 then subtract 5. This means Bus B had 37 at the beginning. **Bus A** must have had **40 passengers** (77 - 37).

51	1	Although a cylinder has three faces in total, two of these are flat (circles) and only **one** is curved.
52	620	Change the first calculation into 5208 ÷ 84 = 62. If the 84 becomes ten times smaller then the 62 must become ten times greater *(compare 100÷10=10 with 100÷1=100)*.
53	315	There are 300 minutes in 5 hours (60 x 5). 0.25 means one-quarter. There are 15 minutes in one quarter of an hour. 300 + 15 = **315**
54	(8,3)	If the mirror line is perfectly diagonal and runs through (0,0) then (1,1) etc., reflections are simple: just swap the numbers. Point A is at (3,8). Its reflection must be at **(8,3)**.
55	2.45 litres	His first four bottles must be one of each in the picture: 300ml+500ml+400ml+250ml. This equals 1450ml. He can buy two more. Two more 500ml is 1000ml. The total is 2450ml, which is **2.45**L.
56	54 cm	Find the ‘tallest’ part of the shape (7cm on the left) and the widest part (20cm across the middle). Use these two measurements like you would with a standard rectangle.The perimeter would be **54cm**.

Test 7 English Answers

1	N	There are no mistakes. The first use of an apostrophe is for possession: the *special* belongs to *tonight*. The second use of an apostrophe is for contraction (*it's = it is*). The waiter has asked a question (*May I...*) so a question mark is appropriate.
2	D	There is **no full stop** before the closing speech mark. The use of a capital for *Sir* is appropriate as it is being used as a title. The comma before *Sir* is also needed, as we must separate the main sentence from the name of the person to whom we are speaking.
3	C	There needs to be a **<u>re</u>opening speech mark** before the word *you*.
4	C	A **comma** is needed to separate the words *challenge* and *Oscar*. The main clause of the sentence is *Oscar climbed to the top diving board*. The introductory clause is *Never one to shy away from a challenge*. Punctuation must separate these clauses.
5	B	*Belfast City <u>H</u>all* is a proper noun phrase, which means that **capital letters** should be used at the beginning of each word (unless they are 'unimportant' words like we find in *The Great Wall <u>of</u> China*).

6	C	*They couldn't hear each other speaking* **due** *<u>to</u> the volume of the music*. *Due* is the best word is because it is followed by the word *to*. None of the other options should be followed by *to*.
7	D	*James is the* **taller** *of the <u>two</u> brothers*. We must use the comparative *tall<u>er</u>* rather than the superlative *tallest*. Only use words like 'tallest', 'best', 'happiest' when the comparison is between <u>more than two</u> nouns.
8	B	*I had better receive a* **response** *to my question soon. Answer* and *explanation* begin with vowels, so the preceding word would have had to be *an*. The word *replies* is plural, so would also not be preceded by the word *a*. We do not say '*<u>a</u> feedback*'.
9	E	*Sasha <u>had</u>* **written** *dozens of letters to the mayor.* We must use a past partciple. The preceding word is *had*. Compare these two correct phrases: Sasha **wrote** / Sasha **<u>had</u> written**.
10	E	*'You can borrow my car* **on the condition that** *you fill it up with petrol afterwards.'* We need a phrase that matches the meaning of the word *if*.

11	B	The correct spelling is **su<u>r</u>prise**. There is an r in the first syllable as well as the second. Use the following sentence to emphasise this need for a *sur-* prefix: *The <u>sur</u>geon was <u>sur</u>prised that the <u>sur</u>vivor <u>sur</u>vived.*
12	A	The correct spelling is **o<u>f</u>**. One helpful way to distinguish *off* from *of* is to pronounce 'of' like this: '<u>uv</u>'. This is what we naturally say when not paying attention to the word. *Because of the heavy rain*... can sound exactly like *Because* 'uv' *the heavy rain*...
13	D	The correct spelling is **l<u>ei</u>sure**. Do not use the *i before e except after c* rule, as the two letters do not make the long ee sound. Focus on the letter i then the word *sure*: '*I go to the le<u>isure</u> centre because <u>I sure</u> like swimming*'.
14	B	The correct spelling is **stre<u>t</u>ching**. It is more common to see *-tch* after a <u>short</u> vowel sound (e.g. *c<u>a</u>tch*, *f<u>e</u>tch*, *cl<u>u</u>tch*) but to see *-ch* after a <u>long</u> vowel sound (e.g. *b<u>ea</u>ch*, *c<u>oa</u>ch*, *p<u>ou</u>ch*).
15	A	The correct spelling is **<u>grate</u>ful**. When someone is *grateful* they are not 'full of great'! They are full of *<u>grat</u>itude*.

16	C	The main character learned that it would prefer to be at home. It had a thirst to see the 'outside world' but did not enjoy the experience in the end. **There's no place like home** is the most suitable expression to convey this idea.
17	A	*See line 1: It was one of those* **dismal** [dreary / dull] *nights...*
18	E	The phrases *thumps and bumps*, *clatters and crashes* and *wear and tear* are telling us how **tired** the bowling ball is becoming. The final sentence of the paragraph then sets the story in motion: it longs for a *better life* (it **starts dreaming of change**).
19	E	The clue is in lines 12-13 (the beginning of the sentence): *When <u>all eyes were averted to another part of the hall</u>*.... The escape was **unnoticed**.
20	D	The character did not encounter **broken glass** during its adventure. *Slick* streets (line 17) describes *slippery surfaces*. *Stray dogs* (line 22) are *homeless animals*. The *traffic collision* in on line 22. The *toppling tree* is on line 23.
21	A	This is **alliteration**: the repeated use of similar starting sounds: *<u>e</u>ach of them <u>e</u>mitting <u>e</u>xclamations of <u>e</u>nthused <u>e</u>xhilaration*. Ususally, alliteration uses consonant sounds, but vowels can create the same effect (e.g. *the <u>o</u>pen <u>o</u>ceans <u>o</u>verflowed*).
22	D	Lines 27-28: *Its weight had further diminished* [it was **physically lighter**] *from the lengthy and arduous journey, but its heart* [emotions] *felt much, much* **heavier**.

23	protagonist	The word is on line 11. A **protagonist** is the main character in a story.
24	countenance	This word appears on the same line as *face*. Although it is not a word that children are expected to be familiar with, they should be able to swap tricky words for *face* and work out which one makes sense.
25	Front Pin	See lines 44 & 45. The bowling ball was never given a name. Kids should be looking a character with capital letters.
26	pronoun	The word *it* replaces the noun *ball* (or *bowling ball*). This makes it a **pronoun**. Similarly, we use the pronouns I, me, you, he, she, they, we, etc. to replace names/nouns.
27	Tuesday	The main character spends the night and early morning outdoors, then returns home and falls asleep. Look at line 47: *That evening...* It is therefore still the next day: **Tuesday**.
28	preposition	A **preposition** is a 'position' word (e.g. *on*, *under*, *through*, *between*). It is usually followed by a noun to which it is connected (e.g. line 13: ...*from* its *lane*; line 24: ...*past* deserted *parks*).

29	D	Seven halves means 7 x 0.5, which is **3.5**.
30	D	The **square** comes after the pentagon (see Shapes 3 & 4). So, as the pentagon is Shape 8, the square must follow it as Shape 9.
31	B	The three measurements must multiply to make 168. In other words, 3 x 8 x ? = 168. The fastest way to solve this is to divide 168 by 8, then divide that answer by 3 (or divide by 3 then by 8). The answer is **7cm**.
32	C	It is best to write out the number in the question first. The word *thousand* is a helpful place to position a comma. Five hundred and sixteen...comma... then thirty (but with three digits) = 516,030. Then add one: **516,031**.
33	D	If the square's perimeter is 24cm then each of its sides must measure 6cm (24 ÷ 4). This means its area is 36cm² (6x6). The rectangle has the same area (36cm²), and one of its sides is 4cm. 4 x ? = 36. The missing length must be **9cm**.
34	E	The times added together make 57 minutes. There are nine items on the schedule, which means eight one-minute gaps. Add 57 and 8 to make 65 minutes (or one hour and five minutes). The concert will finish at **8.20pm**.
35	D	415 x 8 = 3320. This is the number of grams. Divide by 1000 to get the number of kilograms: **3.32**.
36	A	A **cone** has two faces, one of which is curved (i.e. not flat). Cylinders (3) and spheres (1) have an odd number of faces. Square-based pyramids and cubes do not have any curved faces.
37	B	6.3 - 0.7 = 5.6. 5.6 x 4 = **22.4**.
38	C	Sequence A will lead to a tree on the second instruction. Sequence B will lead to a tree on the final instruction. Sequence D will lead to a tree on the third instruction. Sequence E will lead to a tree on the final instruction.
39	C	Convert each fraction into twelfths by making equivalent fractions. This list will say 7/12, 9/12, **10/12**, 8/12 and 6/12. The largest of these is **10/12**, which means that **5/6** is the answer.
40	D	57 is not a prime number (add its digits together to make 12, which is a trick for discovering multiples of three). The largest prime number in the list is **47**.
41	D	Round 386 to its nearest hundred. Round 61 to its nearest ten. The best estimation is therefore 400 x 60, which is **24,000** (the same as 4 x 6 with three zeros).
42	A	Count the cubes in the front layer. There are 12. The back layer is almost the same. It just has one extra cube on the top-right. So, there are 12 in the front and 13 in the back, making **25 cubes** altogether.
43	C	Sketch the smaller triangles inside the larger one. Two will fit across the base and reach halfway up. One sits on top. Another one can be placed 'upside-down' in the gap. This means **four** will fit altogether.
44	A	100 minutes is 1 hour 40 minutes. Perhaps it's easier to subtract two hours then add 20 minutes back on. This takes us from 02:00 to 00:00, then **00:20**.
45	C	Bottle A has 10,000ml. 10% of this is 1000. Subtract this from 10,000 to get 9,000 for Bottle B. 10% of this is 900. Subtract this from 9,000 to get 8,100 for Bottle C. 10% of this is 810. Subtract this from 8,100 to get **7,290ml** for Bottle D.
46	D	75 is three times greater than 25. This is half of Joanna's number, so multiply 75 by two to get her number: **150**.
47	D	Each interval on the vertical axis is worth 5%. Timothy's six scores were 60%, 70%, 85%, 75%, 55% and 75%. Find the mean by adding these numbers then dividing by six. They add to make 420. 420 ÷ 6 = **70**.
48	A	The three angles must add to make 180˚. The two given angles add to make 107˚. The remaining angle must be **73˚**.
49	D	The trainers cost less than the jacket. Divide by 3 then multiply by 4 to get the jacket's cost. 48 ÷ 3 = 16. 16 x 4 = **64**.
50	A	Kids don't need to work out all the cube numbers. As long as they know that 5^3 produces the first three-digit cube number (125) and that 10^3 is the next one that doesn't have three digits (1000), they can work out that there are **five** numbers altogether.

51	270	He performs 10 on Monday, 20 on Tuesday, 40 on Wednesday and 80 on Thursday. This makes 150 so far. It should be 160 on Friday but 75% (¾) of this is 120. Add 120 to 150.
52	66 cm	The two triangular faces have a combined perimeter of 30cm. This leaves us with three edges that each measure 12cm. 30+12+12+12 (or 30+36).
53	0.71	Point A is at 0.1. Point B is at 0.25. Point C is at 0.36. These add to make **0.71**.
54	4 cm	Split it into two rectangles with a horizontal line. The top rectangle has an area of 30cm². The lower rectangle must have an area of 12cm². Its horizontal length is 3cm. The other must be **4cm**.
55	128 minutes	Katie will be catching the bus at 5.53pm, which arrives in Marville at 7.28pm. But don't use 5.53pm in the calculation. She arrives at the stop at 5.20pm, which is **2 hours 8 minutes** before 7.28pm.
56	45	Change 5.5m into 550cm. Divide this by 12. Any remainder (or decimal) should be ignored as Jonah can't make a full length of string with this 'leftover'.

TEST 8 ANSWERS >>>

Test 8 English Answers

1	D	The closing speech mark should be **after** the question mark.
2	A	_African_ must begin with a **capital letter** as it is a proper adjective.
3	B	A **comma** is needed to separate the names *Olivia* and *Lucy*. Otherwise the sentence looks confusing: is there a girl called *Olivia Lucy*? Even if so, there would be no flow to the sentence. It only makes sense if these are different girls.
4	A	Look at the pair of commas in question 2. A **pair of dashes** can perform the same job: separating a dependent clause from the main sentence: *That old oak tree – the one by the river – has been standing there for over a century*.
5	C	A **possessive apostrophe** is needed for the word *kids*. If only one *kid* is being spoken about, it would be *kid's favourite ice-cream flavours*. Alternatively, it would be *kids' favourite ice-cream flavours* if more than one child is being spoken about.

6	A	*He communicated his message to the audience* **through** *an interpreter*. The meaning of *interpreter* is key. The message can only reach the audience *through* the interpreter's work.
7	D	*Thomas* **has been waiting** *all day but the postman still hasn't arrived*. The use of *hasn't* informs us that the sentence is set in the present tense. As the situation is still ongoing (waiting for the postman) we use *has been waiting*.
8	A	*The Titanic* **sank** *in April, 1912*. A simple past tense is needed: *sank*. Only use *sunk* if it is preceded by a verb like *has*, *have*, etc. (e.g. The Titanic **sank** / The Titanic **has sunk**).
9	E	*'I'd rather you didn't tell me now,* **unless** *it's really, really important.' Unless* is used to state certain conditions. Only use *except* when we know the full range of other possibilities (e.g. Everyone can go *except* Jack / Jack can't go *unless* he behaves).
10	C	*Of all the tasks today, organising the cupboard is definitely the* **least** *important.* A superlative adjective has to be used because we may assume there are more than two tasks to choose from.

11	D	The correct spelling is **cousins**. *Cousin* is an unusual spelling as it comes directly from an Old French word. The *cou-* at the beginning has the same sound as the beginning of *couple*.
12	B	The correct spelling is **fasten**. It comes from a definition of *fast* that means 'firm, fixed and secure' (we still use expressions like 'hold fast').
13	A	The correct spelling is **careful**. The *-ful* suffix for adjectives should never have a double L (e.g. *beautiful*, *wonderful*, *awful*).
14	D	The correct spelling is **beginning**. There is a double n. This is because it is the syllable that is being stressed ('be-GINN-ing').
15	B	The correct spelling is **appearance**. Most words that start with the letter a will have an *-ance* ending rather than an *-ence* ending (e.g. *abundance*, *acceptance*, *assistance*, *attendance*, *ambulance*). *Audience* and *absence* are among the few exceptions.

16	E	It is not option A (see lines 26-27). It is not option B (see lines 10-11). It is not option C (see lines 25-26). It is not option D (see lines 29-32).
17	B	See lines 19-21: *By consuming* **algae and aquatic plants that would otherwise overgrow** *and create issues, they contribute to the maintenance of the ecosystems and rivers in our area.*
18	E	Options A, B, C and D are named in lines 6-7. **Canals** are not named in the passage.
19	C	See lines 17-18. Turtles depend on their *wits* (which means **intelligence**).
20	D	'...*do your research*...' In this phrase, *research* is a thing**,** so must be a **noun**. The clue is in the preceding word: *your*. If it belongs to *you*, it must be a thing. However, in a phrase like '*I need to research some facts*', the word would be a verb.
21	A	See lines 12-13. While the shells are made of bone, the *outer layer* has a coat of **keratin**.
22	A	See **line 5**: ...*there is an extensive variety* [wide range] *of turtles*...

23	charming	It is an opinion to say that turtles are **charming**. Not everyone agrees!
24	beach	Lines 32-33: ...*If you see a turtle laying eggs* **on the beach**...
25	plastron	See lines 11-12: *They have two components to their shells: the upper part, known as the carapace, and the lower part, known as the* **plastron**.
26	Green sea turtle	See line 16: **Green sea turtles** *have been known to live longer than 100 years.*
27	deplete	See line 27: ... *which can further* **deplete** [reduce] *turtle populations.*
28	31	The hyphenated word is **single-use**. The line number must be given,

29	B	The kite is 'on its side' (kids normally see kites as having the two longer sides extend towards the bottom). A kite must have two shorter sides of equal length that meet, and two longer sides of equal length that meet. It will also have one line of symmetry.
30	D	A large cone with two extra toppings costs £3.45 (£2.85 + 60p). A medium tub with one extra topping costs £3.25 (£2.95 + 30p). The difference between these prices is 20p, or **£0.20**.
31	D	An isosceles triangle cannot have a 60˚ angle (the other two angles would also have to be 60˚, making it an equilateral), so statement A is not true. **Statement D** is true: an isosceles can have a right angle (the angles would be 90˚, 45˚ and 45˚).
32	A	60% means ⁶⁰⁄₁₀₀. This can be simplified to make ⁶⁄₁₀ (divide both numerator and denominator by ten). This can be simplified again to make ³⁄₅ (divide both numerator and denominator by two).
33	E	**Not all the numbers are even** (the first number is 1) so option E is the false statement. The completed sequence will be 1, 2, 4, 8, 16, 32, 64, 128. The first seven numbers are factors of 128. 1, 4, 16 and 64 are square numbers. 1, 8 and 64 are cube numbers.
34	B	A cuboid has 6 faces. A triangular-based pyramid has 6 edges. A triangular prism has 6 vertices. A square-based pyramid has 5 faces. These numbers add to make **23**.
35	B	The digit 7 is in the **ten thousands** place (the order of whole-number places, going from right to left, is units, tens, hundreds, thousands, ten thousands, hundred thousands, millions).
36	D	Rectangle A has a perimeter of 21cm. Rectangle B is 22cm. Rectangle C is 22cm. **Rectangle D is 23cm**. Rectangle E is 19cm. However, it doesn't really matter. All we need to do is add one of each measurement – the largest answer wins!
37	D	2½ hours = 150 minutes, so create a fraction and then simplify it. The fraction is ²⁵⁄₁₅₀, which can be simplified to make ⁵⁄₃₀, which can be simplified again to make **⅙**.
38	B	If Zara answered all 8 questions correctly, she would be on step 24 (3x8). If she had 7 right and 1 wrong, she would be on step 20 (3x7 then -1). This means it's impossible to be on **step 22**. Keep counting back from 20 in fours - the others are possible.
39	C	Convert all measurements into centimetres. The list would read like this: 47.5cm, 5000cm, 48cm, 50cm, 47.25cm. The third longest measurement is 48cm, which is **480mm**.
40	A	The first two prime numbers are 2 and 3, which have a sum of **5**. A prime number must have exactly two factors, which is why 1 is not a prime number.
41	C	The number must be greater than 10 because 28 x 10 = 280. It must be less than 20 because double 280 is 560. This only leaves **12** as a possibility. There is no need to perform any trickier calculations.
42	A	Angles next to each other in a parallelogram add to make 180˚. The bottom-right angle must be 135˚. This 135˚ angle sits next to the bottom-left angle on the triangle. They make 180˚ (a straight line), so it's 45˚. We can then work out Angle A: **45˚** also.
43	E	20% is the same as ⅕. One-fifth of £85 is £17. Subtract this from the price to get **£68**.
44	A	These circles are arranged according to triangular numbers. So far, the sequence is 1 *(+2)* 3 *(+3)* 6 *(+4)* 10. We add five circles to find the next number: **15**.
45	D	Find ¾ of one of the three measurements then multiply it by the other two measurements. For example, ¾ of 8cm is 6cm, so 6 x 20 x 10 = **1200**.
46	C	One cycle of exercises (including rests) lasts 12 minutes. So, to find what Lucy was doing after 20 minutes, look for what she was doing after 8 minutes (20 - 12). It will be the same. It was the **sit-ups**.
47	A	£6.99 x 7 is solved quickest by doing £7 x 7 then subtract 7p: £48.93. This means £1.07 change from a £50 note. This can be made with **three coins**: £1, 5p and 2p.
48	D	Compare the two calculations. The first number has become ten times smaller, the second number has become ten times greater. This means the answer stays the same: **6015**.
49	B	A square with an area of 100cm² must have sides that measure 10cm each (10x10 to find the area). The perimeter must therefore be **40cm** (10+10+10+10).
50	E	115 minutes is five minutes less than two hours. Two hours after 11.45am is 1.45pm. Subtract five minutes to get to 1.40pm. As a 24-hour time, this is **13:40**.

51	1	**1** is a square number (1x1) and a cube number (1x1x1).
52	60	This is a matter of how many times 1.5 fits into 90 (or how many times 1500 fits into 90000). 1.5 fits into 9 six times, then make this ten times greater to fit into 90. 6 x 10 = **60**
53	17	Add the pictures of a ball, including the half and quarters. There are 12 of them. As there were 48 goals in the tournament, each ball must be worth 4 goals (48 ÷ 12). 4¼ x 4 = **17**.
54	108°	Including the zero, there are ten evenly-spaced numbers on the board. As there are 360˚ in a full circle, each section must be worth 36˚ (360˚ ÷ 10). The angle to reach number 3 must be 36˚ x 3.
55	6	-18 + 60 = 42 42 ÷ 7 = **6**
56	£2.25	1.25 litres (or 1250ml) is 2½ times greater than 500ml, so multiply 90p by 2.5 (or add 90p, 90p and 45p).

TEST 9 ANSWERS >>>

Test 9 English Answers

1	N	There are no mistakes. The comma has been correctly placed so as to separate the introductory clause (*Every time her friends came around*) from the main clause (*something would end up getting broken*).
2	C	The **comma** should be before the closing speech mark, not after it.
3	C	*Perimeters* is a simple plural. **It should not have an apostrophe**. The *perimeters* are not said to possess anything in the sentence. Furthermore, neither *squares* nor *rectangles* have an apostrophe, so this is also a clue.
4	B	Look at sentence 1; this gives us a clue as to structure. An introductory clause is given, followed by the main clause. These must be separated by a **comma**. *Under the microscope**,** the ant's tiny features...*
5	D	Clara is not asking a question. She is stating something about the warning she gave. **There should not be a question mark**. Even though the sentence is long and complex, the only punctuation error is straightforward.

6	C	*The film was* **so** *exciting that we watched it again the next day.* The other four options can certainly be used to describe *exciting*, but they would not work with the second half of the sentence.
7	A	*She couldn't decide* **whether** *to go to the party or stay home and rest. Whether* is the word we use when we consider choosing one thing or the other.
8	D	**More** *people are choosing to work from home than those who prefer to come to the office.* Rephrase the sentence to see why this works...*The number of people who are choosing to work from home is more than the number of people who come to the office.*
9	C	*The winners* **will be announced** *after the teachers have looked at each piece of art.* We are looking for a future tense as it's clear that the teachers haven't yet looked at each piece of art.
10	D	**Beneath** *her confident appearance, Katy was feeling extremely nervous.* We must choose a preposition that gives the idea of Katy's feelings being 'hidden behind' her outward appearance. *Inside* may be technically hidden, but is not the same as 'behind'.

11	C	The correct spelling is **caught**. Along with *taught*, this is an unusual past tense, as words that sound like this typically have an *-ought* ending (e.g. *thought*, *bought*, *sought*).
12	A	The correct spelling is **guaranteed**. Many kids remember that there is a letter u somewhere. However, most don't remember that it comes straight after the letter g. We see this in words like *guard* as well.
13	N	There are no mistakes. *Gruesome* does not drop its first e. Watch out though: *true* drops its e when we spell *truly*; *argue* drops its e when we spell *argument*.
14	D	The correct spelling is **their**. It is a homphone of *there*. *Their* means 'belonging to them'. *There* refers to a place (e.g. we are going *there* tomorrow).
15	B	The correct spelling is **inventor**. There is no easy way to know whether words have *-er*, *-or* or *-ar* suffixes. However, many jobs tend to have an *-or* suffix (*actor*, *doctor*, *professor*, *inspector*).

16	D	Count the syllables as you read out the first three lines in any verse. **They each have eight syllables**. Only the fourth line in each verse breaks this pattern.
17	B	Although pain is suggested in the first three verses, the poet's mouth is soon numb, so option E is not the best summary. As it was a bad experience for him, **unpleasant** is the best summary. He was glad to escape (line 29).
18	A	The poet assumes that the dentist planned to **stop him from eating sweets**. The *plan* is only mentioned in the final verse (line 48) and is in the context of the poet not getting the sweets that he wanted.
19	C	See lines 17-20. The *toys* turn out to include a dental drill!
20	E	See **line 29**: *I fled the prison* is a metaphor. The poet was not literally in a prison, but compares the dental experience to being in jail.
21	C	See **line 8:** *Instead, I shrieked*. 'Shrieking' is making a loud high-pitched cry or sound. This is the noise that did not distract the dentist. The poet later makes the sound *num num num num* (line 42) but this is not in the context of noises made at the surgery.
22	E	Options A, B and C should sound too bizarre. Option D is unlikely as he hadn't yet bought the sweets. Option E makes most sense as the poet's mouth is numb and he, therefore, may be having **difficulty in forming words**.

23	agape	Line 6: *My mouth,* **agape** [wide open]...
24	frozen	*My mouth was froze* suits the rhythm of the poem but, grammatically, a participle should be used: *my mouth was* **frozen**.
25	sorry tale	See line 45: *That* **sorry tale** [unfortunate story] *should teach you well...*
26	43	The adjective is *red*. The poet was red because of his embarrassment. The confectioner is red because of his anger.
27	dismay	Although *regret* is often used as a verb (e.g. I regret that), it can also be a noun (e.g. He was full of regret). See line 37: *But realised to my* **dismay** [my regret]...
28	adjective	*Toothy* describes the noun *grin*. *Sharp* describes the noun *tools*. *Strong* describes the noun *teeth*. *Odd* describes the noun *gums*. It is an **adjective's** job to describe nouns.

Test 9 Maths Answers

29	D	£471 ÷ 4 = **£117.75**. Using written division, children are likely to get <u>117 remainder 3</u> as the answer, but this digit 3 cannot be directly converted into decimal form. They must keep using written division in the decimal places (i.e. £471.<u>00</u> ÷ 4).
30	C	The two angles must add to make 180˚. If one number is double another, you can divide their total by <u>three</u> to find the smaller of the two numbers. Angle B must be 60˚ (180˚ ÷ 3). This means Angle A is **120˚**.
31	E	As each percentage value <u>and</u> each number ends with zero, there is a trick we can use: ignore the zeros and multiply. Option A is 2x11; option B is 3x7; option C is 4x6; option D is 5x5; **option E is 9x3**, which is <u>27</u>.
32	B	Rather than guessing (or drawing) what the pattern will look like, or even writing numbers, first study for other patterns in the layout. Pattern 1 contains 1^2 + 1. Pattern 2 contains 2^2 + 2. Pattern 3 contains 3^2 + 3. Pattern <u>5</u> (<u>not 4</u>) will be 5^2 + 5, which is **30**.
33	E	There are six faces on the cuboid. Two of them will measure 6cm by 3cm (18cm² x 2); two will measure 6cm by 4cm (24cm² x 2); two will measure 4cm by 3cm (12cm² x 2). Collectively, these areas add to made **108cm²**.
34	D	Seven tens = 70. Seven tenths = 0.7. <u>Seventy hundredths also equal 0.7</u> (think of the digits of 70 finishing in the hundredths place: 0.70). 70 + 0.7 + 0.7 = **71.4**.
35	C	The length of one side on Square B is 12cm. Its area must be 144cm² (12x12), which is **nine times greater** than the area of Square A: 16cm² (4x4). Square B is not three times greater than Square A – it is 'three squared' times greater.
36	B	There are 1 hour and 10 minutes until midnight (which is <u>70 minutes</u>), then another <u>30 minutes</u> until 00:30. This makes **100 minutes** in total.
37	E	5 + 3.9 + 3.5 + 3.5 = 15.9. 21 - 15.9 = **5.1**.
38	B	Each interval on the bar chart is worth 2 million. Argentina + Chile = 58 million. **Canada + Peru = 70 million**. Chile + Nepal = 44 million. Canada + Argentina = 78 million. Nepal + Peru = 62 million.
39	C	One small sharpener must weigh 6g (72g ÷ 12). 25 of these weigh <u>150</u>g (6g x 25). 25 large sharpeners weigh <u>275</u>g (11g x 25). 275g - 150g = **125g**
40	A	Perpendicular sides will meet at a right angle (90˚). There can never be a right angle in an **equilateral triangle**. Each of its angles must measure 60˚.
41	B	One-fifth is the same as <u>two-tenths</u>. The first digit after the decimal point is the tenths. 2⁄10 = **0.2**.
42	D	The factors of 36 are 1 and 36, 2 and 18, 3 and 12, 4 and 9, and 6. This makes **nine factors** altogether.
43	C	A **triangular prism** will have one triangle at the front and one at the back. These two triangles will be connected by three rectangles, or three squares.
44	B	It will be **8 times greater** because this is the answer to 48 ÷ 6. To test this out, we could use a sample number. 1 would be simplest! 1 x 48 = 48, then 48 ÷ 6 = 8. We started with 1, finished with 8 (which is clearly 8 times greater).
45	A	A quarter-turn is a right angle (90˚). Three quarter-turns from east will bring us through south, then west then finally to **north**.
46	B	If the perimeter of an equilateral triangle is 24cm, then each side must measure 8cm. The arrow at the bottom of the picture is 2cm shorter than the base of a large triangle, meaning that the small triangle's side is 2cm. Its perimeter must be **6cm**.
47	A	From 5th April, there are 25 days remaining in the month, then 6 more days to get to Bobby's birthday, meaning 31 days altogether. 31 days is 4 weeks and <u>3 days</u>. 3 days after Wednesday is **Saturday**.
48	C	30% is 3⁄10. 4.5kg is 4500g. Find three-tenths of 4500 by dividing by ten then multiplying by three. 4500 ÷ 10 = 450. 450 x 3 = **1350**.
49	B	Maisie's total score is 28 (1+1+1+1+3+3+4+4+4+6). 28 ÷ 10 (the total number of rolls) = **2.8**.
50	A	If Isla's number is three times greater than Maggie's then her number is three-<u>quarters</u> of the total, and Maggie's is one-<u>quarter</u> of the total. Divide 84 by <u>four</u> to get Maggie's number: **21** (Isla's number must be 63).

51	60°	The line of symmetry tells us that the top-right angle is also 120˚. Not only this, the two angles in one-half of the shape must add to make 180˚. Subtract 120˚ from 180˚ to get the answer: **60˚**.
52	£9	If Bronagh has one-fifth of her money remaining, then <u>she spent four-fifths</u>. £7.20 must be 4⁄5 of what her mother gave her. £7.20 ÷ 4 = £1.80... £1.80 x 5 = **£9.00**.
53	25%	<u>Two of the eight numbers</u> are prime: 53 and 47. This is the same as one-quarter, which is the same as **25%**.
54	12 days	Find the lowest common multiple of 4 and 6. Do this by counting in 6s until you say a number that is also a multiple of 4. The first number you should say that meets these conditions is **12**.
55	125 cm³	The folded cube will have three identical measurements (length, breath and height) of 5cm. The volume is found by multiplying these three measurements together: 5 x 5 x 5 = **125**.
56	120	0.5 means one-half. If the heart beats every half-second, then it beats <u>twice</u> per second. To find how many beats in one minute, multiply 2 by 60.

Test 10 English Answers

1	B	The **closing speech** mark is missing. It should be written after the comma: *'I've never seen a storm this strong,' shouted the fisherman...*
2	C	The **closing bracket** is missing. The main clause of the sentence is a short one: *The painting hung proudly on her wall*. The extra clause **(***which had taken months to complete***)** must be separate. This can be done with brackets, commas or dashes.
3	D	The **apostrophe** is in the wrong place: it should be *geese's*. The *feathers* belong to the *geese*. Possessive apostrophes must come immediately after the spelling of the name of the owner or owners.
4	N	There are no mistakes in this sentence. It is one single clause that does not need to be broken up with punctuation. The apostrophe in *o'clock* is an apostrophe for contraction (it comes from an archaic expression: *of the clock*).
5	A	*Someone* is not a proper noun; it is a pronoun. **It should not have a capital letter**.

6	E	*Rachel <u>crept</u>* **upstairs***, hoping that the others wouldn't notice.* The first four options are adjectives but we must use an adverb as we are saying something about the verb *crept*. The only possibility among the options is *upstairs*.
7	B	**These** *flowers that I am <u>now</u> holding in my hand were taken from Mrs Newsome's garden.* The flowers are in the speaker's hand. They are 'near', so we use the near demonstrative plural pronoun, *these* (the singular is *this*).
8	A	*She felt most at east* **among** *her group of closest friends.* The sentences is picturing her as part of the group – one of its members.
9	B	*I <u>have</u> never* **run** *a marathon in my life.* We are looking for a past participle, and not a past tense. The past participle of *run* is also *run*! Compare these: I **ran** a marathon last year / I **<u>have</u> run** that marathon before.
10	C	*I thought our performance was great, but Geraldine and Maggie believed that* **theirs** *was the best.* A possessive pronoun is needed: *their* or *theirs*. Use *their* when followed by a noun (e.g. *their dog*). Use *theirs* on other occasions (e.g. *the dog was theirs*).

11	B	The correct spelling is **<u>figure</u>**. The *-ure* ending is quite unusual, considering that we pronounce these final three letters like *-er*, .
12	N	There are no mistakes. The h in *ex<u>h</u>austed* is as silent as the h in *ex<u>h</u>ilarating*. Exaggerate the sound of this letter when saying *ex<u>h</u>austed*: forcing the sound of the h may suitably make one sound exhausted!
13	D	The correct spelling is **<u>ex</u>istence**. Hidden within the spelling of this word is a fact about Roman numerals: *<u>x is ten</u>*.
14	B	The correct spelling is **clothes**. They are made of *<u>cloth</u>*.
15	D	The correct spelling is **unti<u>l</u>**. Just as with *fulfi<u>l</u>*, *aw<u>ful</u>* and *<u>al</u>ways*, there are words that seem like they should have a double L (often because they derive from words with a double L) but only have a single L when combined with other words and prefixes.

16	B	The story is **absurd**. The events (leaving a child with his head stuck between bars for years, bringing a zoo and beach to him, etc.) are too bizarre for real life. It cannot be described as *tense* because there were no imminent threats described.
17	C	The moral (or message) of the story is found in its conclusion. Ewan was trapped because of other people's decisions. It was when he was left to himself that he was able to **push through** and break free from his 'prison'.
18	D	See line 25: **resentful** and *bitter* are very similar in meaning. The clue is lines 24-25 when read together. The classmates were *bitter*; Mrs Thorncroft was *<u>even more</u> resentful*. Even more than what? More than the children...who were *bitter*.
19	A	The **dog** did not appear at Ewan's 'home'. It only appeared when the 'home' had gone (and is inferred to be the same dog as the puppy that had appeared before the story began and the 'home' was built).
20	B	The word is **prestigious**. None of the other options would make sense with this definition. While *upgraded* suggests an improvement in quality, it does not mean that the item is respected and admired.
21	D	See lines 40-42. Note that *<u>the final straw</u>* was when Ewan learned of the plans for a secondary school to be built around him. This was his tipping point (or 'the straw that broke the camel's back').
22	A	*Wonder* is a **noun** in this sentence. It is preceded by the word *a*. Words like a/*an* and *the* always precede a noun (e.g. a pencil; an egg; the school).

23	vainly	See line 14: ...**vainly** *attempting*... It is describing the verb *attempting*. The attempts did not bring about the desired result. They were *fruitless / useless*.
24	abode	See line 35: ...*lorries of sand were transported to his* **abode** [home]...
25	33	Ewan's best friend + 'another kid' + 'another kid' + 'thirty more' = **33**.
26	thud	Onomatopoeia is the use of a word that imitates or suggests the sound that it describes. In this case, **thud** mimics the sound of Ewan landing on the ground after his struggle.
27	publicity	See line 40: ...*the fame and* **publicity** [attention] *became too much for him...*
28	verb	All of these words are what Ewan <u>did</u> in his efforts to reach the dog.

29	D	The shape measures 7cm across the way and 6.5cm down the way. These could be multiplied together to make **45.5cm²**. Alternatively, kids could count each whole square and each half square.
30	C	4.5 x 24 = **108**. This multiplication can be simplified by doubling the first number and halving the second number. It would become 9 x 12, which would be much easier for kids to calculate with.
31	E	We are looking for two-fifths of 1.5km because 2 minutes is two-fifths of 5 minutes. 1500m ÷ 5 = 300m... 300m x 2 = **600m**.
32	C	28 rows of 6 seats is 168 seats (28 x 6). Add 40 first-class seats to get 208. Subtract the 22 empty seats to get **186**.
33	D	Each umbrella is worth two days (see key). October and November each had 19 rainy days. December had 16 rainy days. 19 + 19 + 16 = **54**.
34	B	The hour hand moves through four numbers. There are 30˚ between each number on the clock face (360˚ ÷ 12). 4 x 30˚ = **120˚**.
35	E	Simplify $\frac{15}{20}$ to get ¾. The decimal value is **0.75** (75 hundredths) because ¾ is equal to $\frac{75}{100}$.
36	A	a^2 is equal to 25, which is a square number. Which number gets squared (multiplied by itself) to become 25? It must be **5**.
37	E	A triangular-based pyramid has 6 edges, so we are looking for a shape with 12 edges. A square-based pyramid has 8 edges. A cylinder has 2 edges. A triangular prism has 9 edges. A hexagonal prism has 18 edges. **A cube has 12 edges**.
38	C	It takes five steps (or intervals) to get from 8 to 10. The size of the gap is 2, which must be divided by 5 (2 ÷ 5 is the same as ⅖, which is 0.4). Each interval is worth 0.4, and the arrow is pointing at two intervals after 8. It must be **8.8**.
39	C	This shape has the same perimeter as a full rectangle with a length of 11cm and a width of 8cm. These are the only two measurements that we need. 11+ 8 + 11 + 8 = **38cm**.
40	B	It can't be option A as no ferries sail so late in the day. It can't be option C as ferries only sail at 45 minutes past, 5 past, and 25 past. It can't be option D as weekend times are 10 past, half past, and 50 past. The Sunday time (option E) is too late.
41	C	-11 + 18 = **7**
42	E	Although option A works for the first pair of numbers, it doesn't for the second pair. But option E works: 5 **- 2.5** = 2.5... 2.5 x 8 = 20; 11.5 **- 2.5** = 9...9 x 8 = 72
43	E	Calculate the area of a triangle (2 x 8 then ÷2): 8cm². There are two triangles, making their combined area 16cm². The entire square's area is 64cm² (8x8). Subtract 16cm² from the square's area to get the trapezium's area: **48cm²**.
44	E	25% = ¼. Find the shape with one-quarter shaded. Shape A is $\frac{5}{24}$. Shape B is $\frac{9}{32}$. Shape C is $\frac{7}{24}$. Shape D is $\frac{8}{36}$. None of these fractions match ¼. **Shape E is $\frac{5}{20}$,** which is the same as ¼.
45	D	Only the multiples of three can produce square numbers that are divisible by three, so the answers are 3^2, 6^2, 9^2 and 12^2. There are **four** in total.
46	A	There are 12 edges on a cube and all are the same size. 60cm ÷ 12 = 5cm. The volume will be 5x5x5, which is **125cm³**.
47	D	Write 24.5 but also add a zero for the hundredths place: 24.50. Removing the decimal point will tell us how many hundredths are in the number: **2450**.
48	C	Kids could just test each option out. However, the two measurements will add to make 12cm (half the perimeter). If one number is double the other, then divide by three to get the width: 4cm. The length is double this: **8cm**.
49	A	Lamont won the race: 9.8 seconds. Andre came third: 9.89 seconds. The difference between these times is **0.09 seconds**.
50	E	Instead of one pile of 20p coins and one pile of 10p coins, pair up each 20p coin with a 10p coin. Each pair would be worth 30p. How many times does 30p fit into £2.70. **Nine times**.

51	(4,8)	Connect (3,3) to (6,6): each coordinate goes up by three. The line from (1,5) must be parallel. Add three to each coordinate: **(4,8)**.
52	4624	Perform 7311 - 2687 instead.
53	6	The number of lines of symmetry in a regular shape will match the number of sides it has. *Regular* does not mean 'normal'! It means all sides are the same length and all angles are the same size.
54	21	This is essentially asking us to solve $\frac{12}{18} = \frac{14}{?}$. Simplify $\frac{12}{18}$ to get ⅔. Now the question is ⅔ = $\frac{14}{?}$. The numerator has been multiplied by seven. The denominator of 3 must also do the same.
55	6	Count the squares in each pattern and determine the subtraction sequence. The numbers are 28 *(-7)* 21 *(-6)* 15 and *(-5)* 10. These are triangular numbers. The next step will be to subtract four: **6**.
56	1920 cm³	20 x 12 x 8 = **1920**

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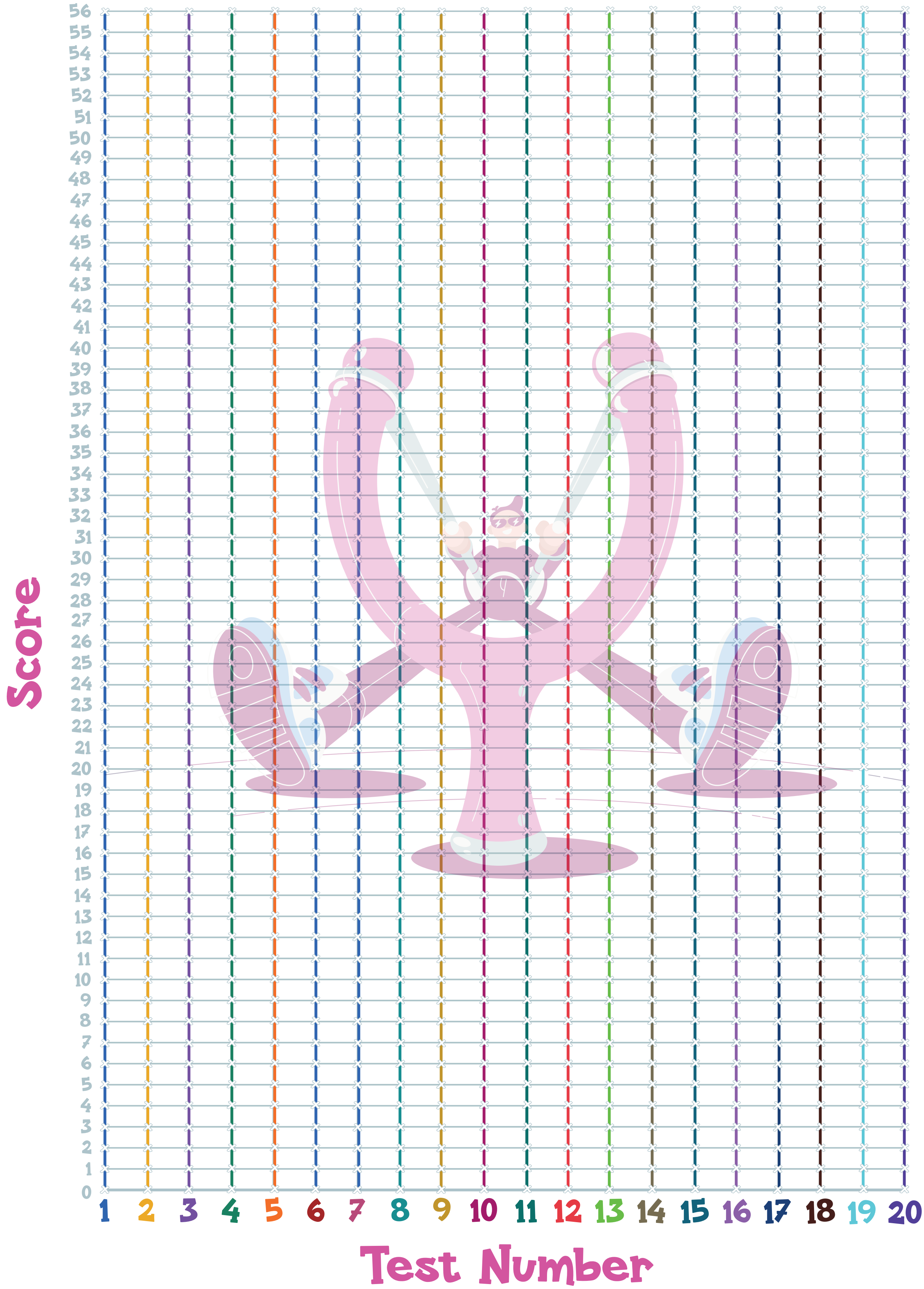

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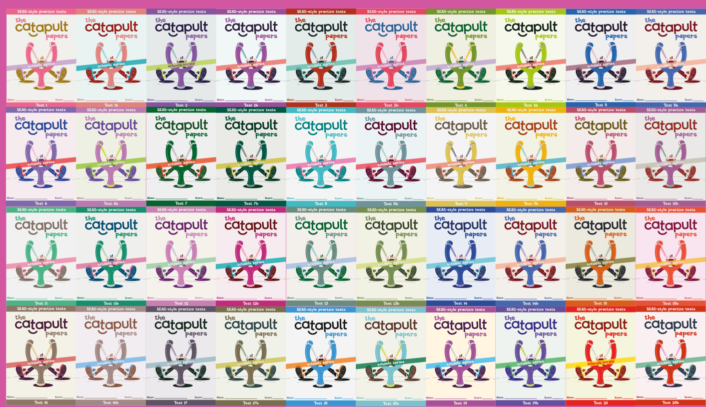

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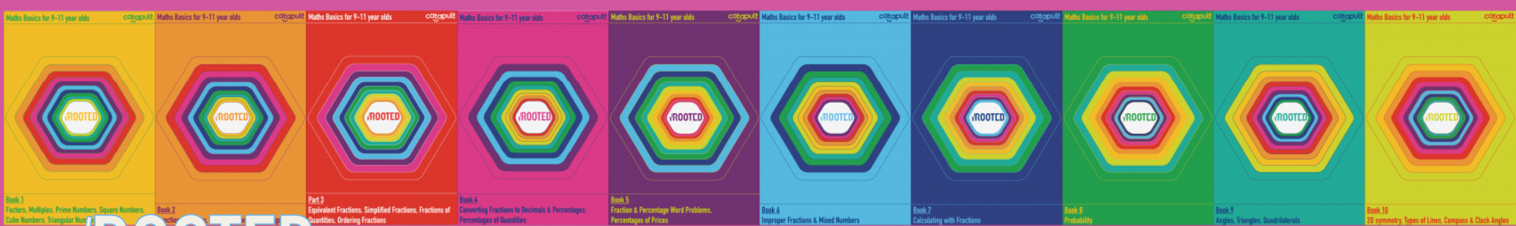

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